

Fitted Models

Loïc Pages

2025-07-01

Seedlings survival

Models selected by AIC

```
ASurvglm11 <- fitme(Survie ~ 1+ bs(Size0Mars,df=4,degree=2) + (Size0Mars|year),
                    family=binomial,
                    data=survdata1,
                    method="PQL/L")

ASurvglm12 <- fitme(Survie ~ 1+ bs(Size0Mars,df=4,degree=2) + (Size0Mars|year)+ (1|Pop),
                    family=binomial,
                    data=survdata1,
                    method="PQL/L")

ASurvglm13 <- fitme(Survie ~ 1+ bs(Size0Mars,df=4,degree=2) + (1|year) + (1|Pop),
                    family=binomial,
                    data=survdata1,
                    method="PQL/L")

ASurvglm14 <- fitme(Survie ~ 1+ bs(Size0Mars,df=4,degree=2) + (1|year),
                    family=binomial,
                    data=survdata1,
                    method="PQL/L")

ASurvglm15 <- fitme(Survie ~ 1+ bs(Size0Mars,df=5,degree=3) + (Size0Mars|year),
                    family=binomial,
                    data=survdata1,
                    method="PQL/L")
```

Models selected by BIC

```
BSurvglm11 <- fitme(Survie ~ 1+ Size0Mars + (1|year) + (1|Pop),
                    family=binomial,
                    data=survdata1,
                    method="PQL/L")

BSurvglm12 <- fitme(Survie ~ 1+ Size0Mars + (1|year),
                    family=binomial,
                    data=survdata1,
                    method="PQL/L")
```

```

BSurvglm13 <- fitme(Survie ~ 1 + Size0Mars + (Size0Mars|year) + (1|Pop),
  family=binomial,
  data=survdata1,
  method="PQL/L")

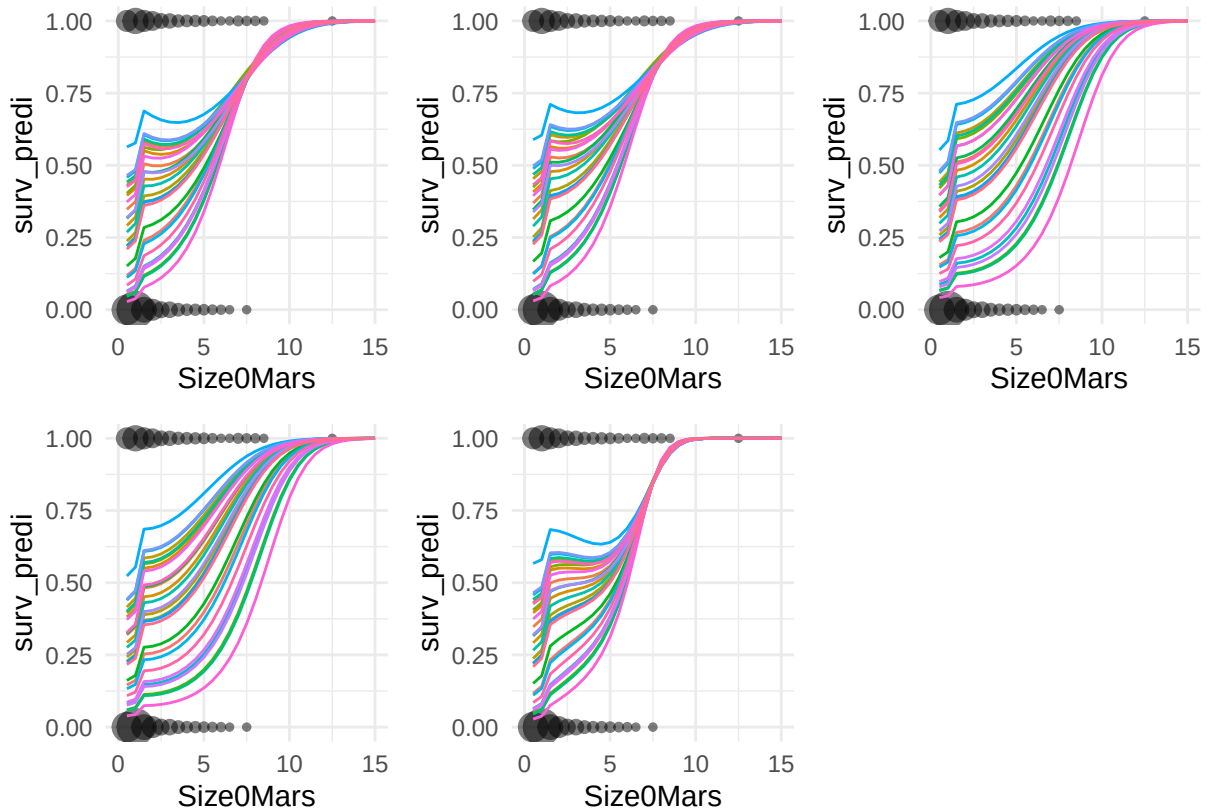
BSurvglm14 <- fitme(Survie ~ 1+ Size0Mars + (Size0Mars|year),
  family=binomial,
  data=survdata1,
  method="PQL/L")

BSurvglm15 <- fitme(Survie ~ 1+ Size0Mars + (1|year) + (Size0Mars|Pop),
  family=binomial,
  data=survdata1,
  method="PQL/L")

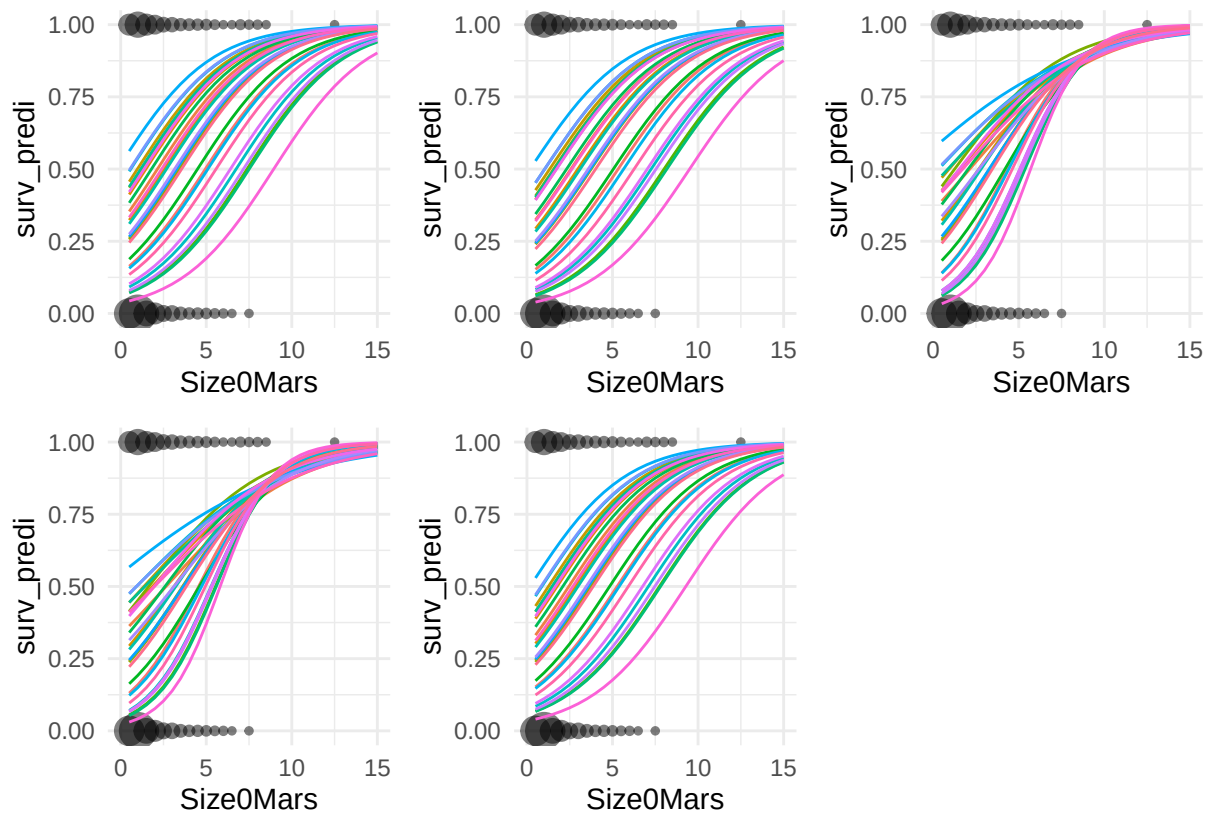
```

As a function of size

See year effect (population fixed)

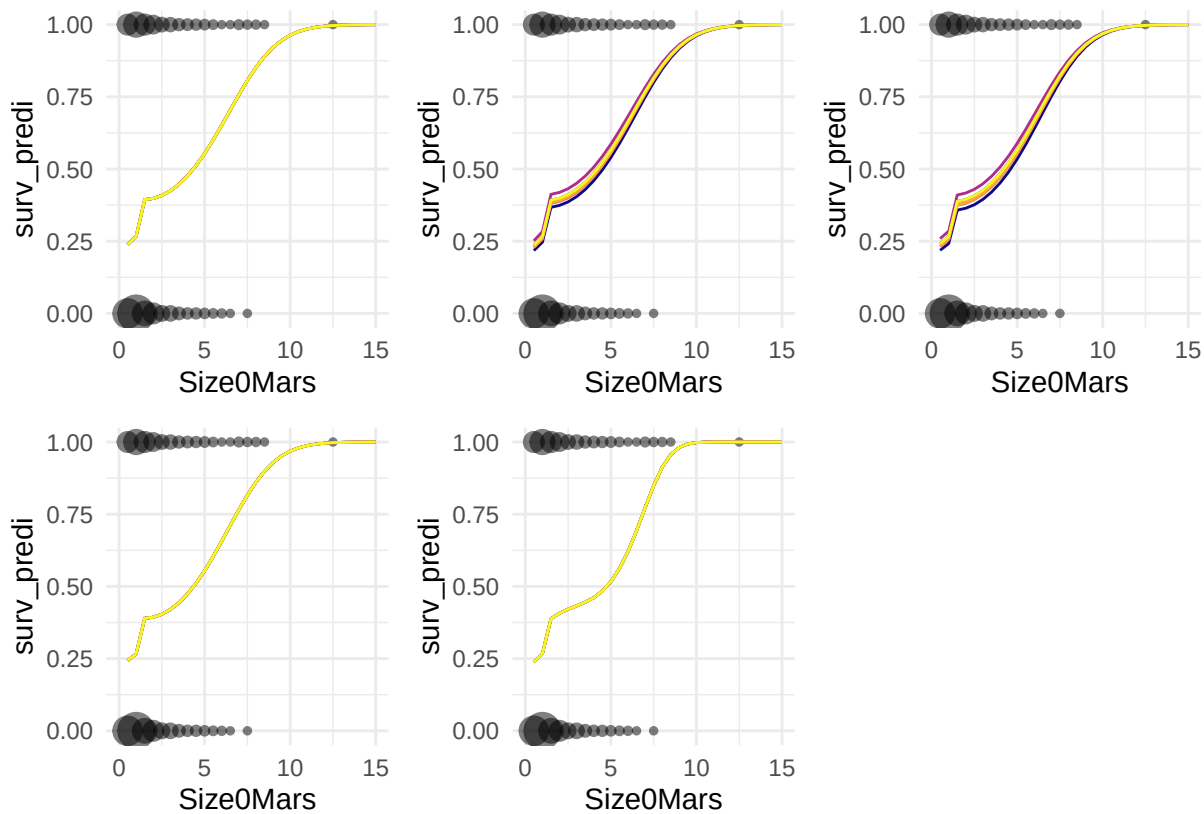


AIC

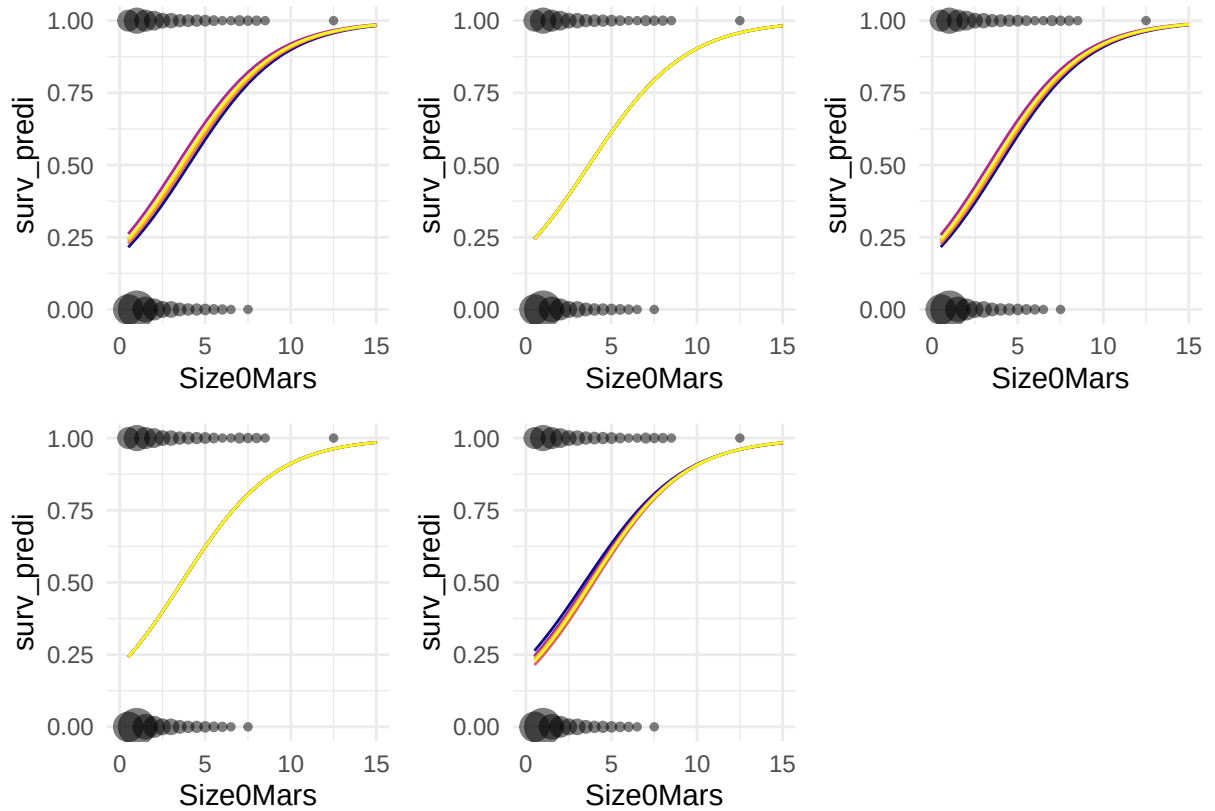


BIC

See population effect (year fixed)



AIC



BIC

Rosette survival

AIC

```
ASurvglm21 <- fitme(Survie ~ 1+ bs(Size0Mars,df=3,degree=2) +
  bs(Age,degree=3,knots=6.5) + (Age|year) + (1|Pop),
  family=binomial,
  data=survdata2,
  method="PQL/L")

ASurvglm22 <- fitme(Survie ~ 1+ bs(Size0Mars,df=3,degree=2) +
  bs(Age,degree=3,knots=6.5) + (1|year) + (1|Pop),
  family=binomial,
  data=survdata2,
  method="PQL/L")

ASurvglm23 <- fitme(Survie ~ 1+ bs(Size0Mars,df=4,degree=2) +
  bs(Age,degree=3,knots=6.5) + (Age|year) + (1|Pop),
  family=binomial,
  data=survdata2,
  method="PQL/L")

ASurvglm24 <- fitme(Survie ~ 1 + poly(Size0Mars,4) + bs(Age,degree=3,knots=6.5)
  + (Age|year) + (1|Pop),
```

```

        family=binomial,
        data=survdata2,
        method="PQL/L")

ASurvglm25 <- fitme(Survie ~ 1+ bs(Size0Mars,df=4,degree=2) +
                    bs(Age,degree=3,knots=6.5) + (1|year) + (1|Pop),
                    family=binomial,
                    data=survdata2,
                    method="PQL/L")

```

BIC

```

BSurvglm21 <- fitme(Survie ~ 1+ bs(Size0Mars,df=3,degree=2) + (Age|year) +
                    (1|Pop),
                    family=binomial,
                    data=survdata2,
                    method="PQL/L")

BSurvglm22 <- fitme(Survie ~ 1+ poly(Size0Mars,3) + (Age|year) + (1|Pop),
                    family=binomial,
                    data=survdata2,
                    method="PQL/L")

BSurvglm23 <- fitme(Survie ~ 1 + bs(Size0Mars,df=3,degree=2) + (Age|year) +
                    (Size0Mars|Pop),
                    family=binomial,
                    data=survdata2,
                    method="PQL/L")

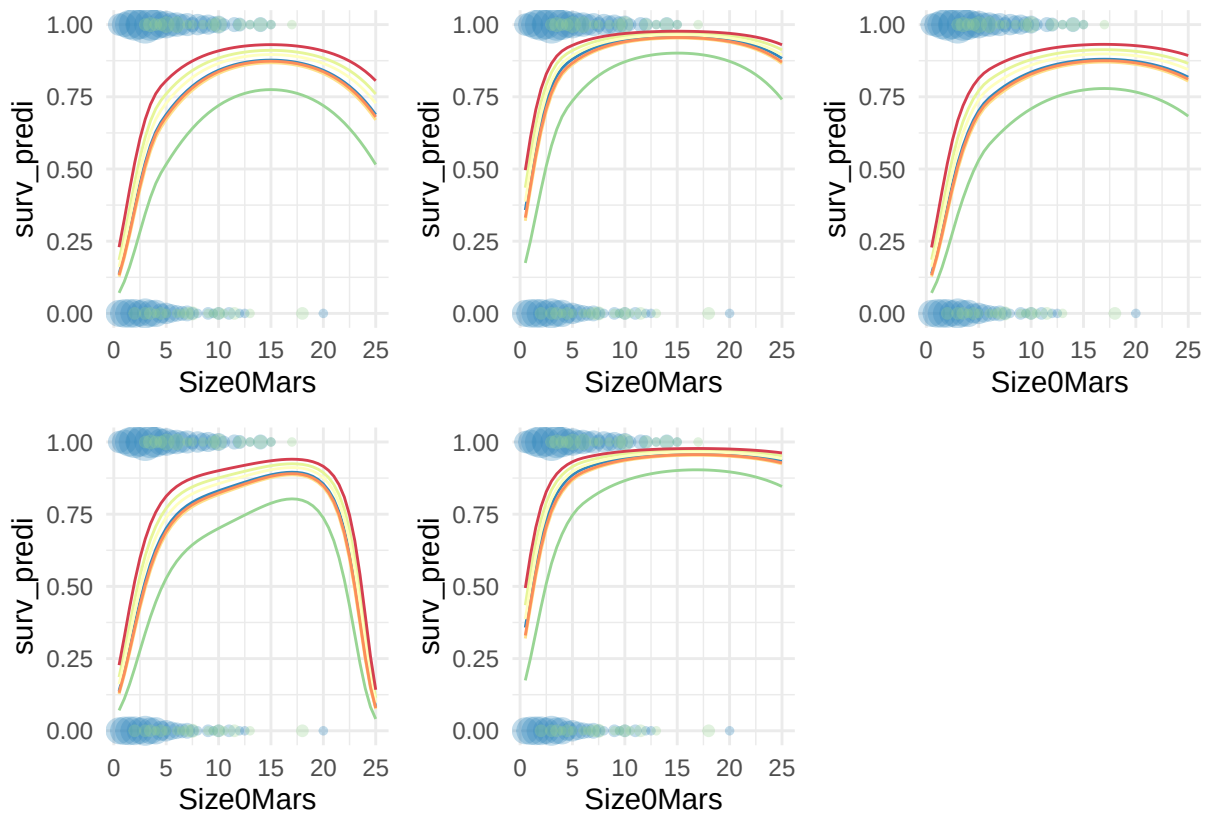
BSurvglm24 <- fitme(Survie ~ 1+ bs(Size0Mars,df=3,degree=2) + (Age|year) +
                    (Age|Pop),
                    family=binomial,
                    data=survdata2,
                    method="PQL/L")

BSurvglm25 <- fitme(Survie ~ 1+ poly(Size0Mars,2) + (Age|year) + (1|Pop),
                    family=binomial,
                    data=survdata2,
                    method="PQL/L")

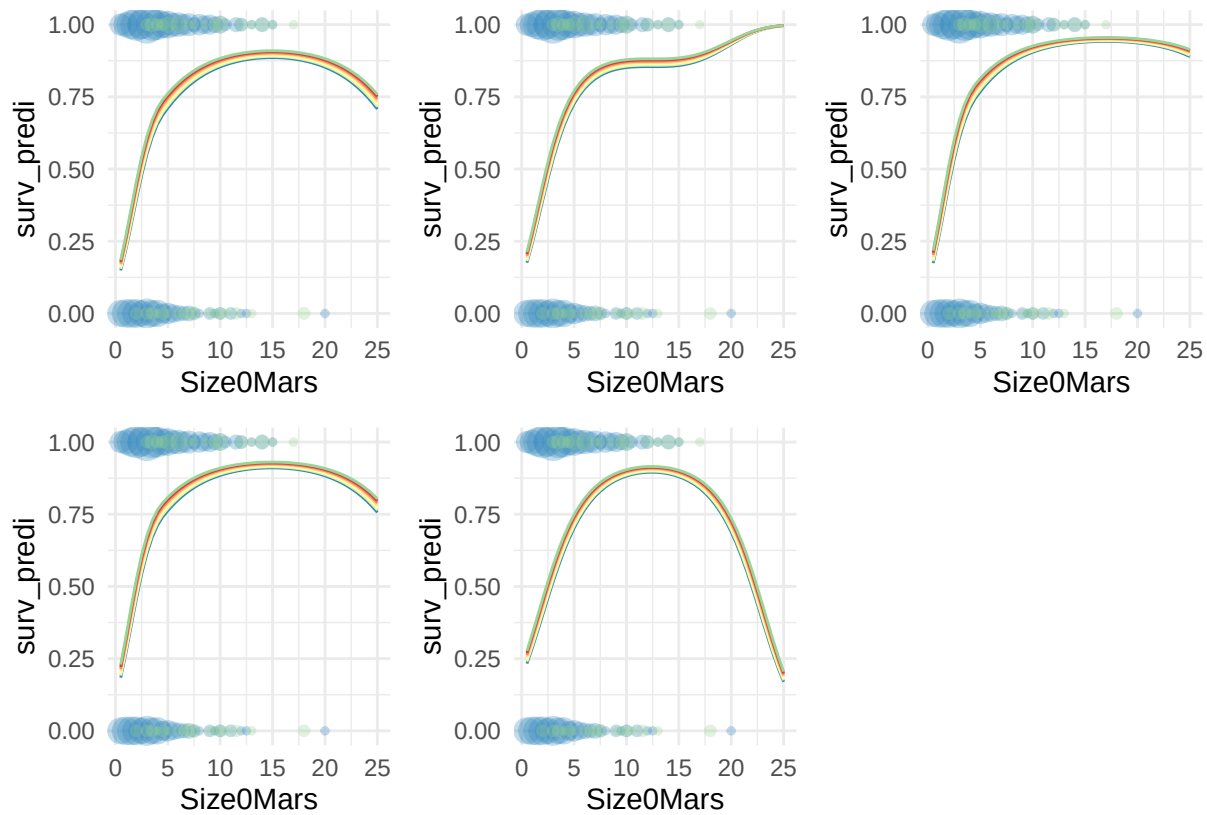
```

As a function of size

See effect of age (population and year fixed)

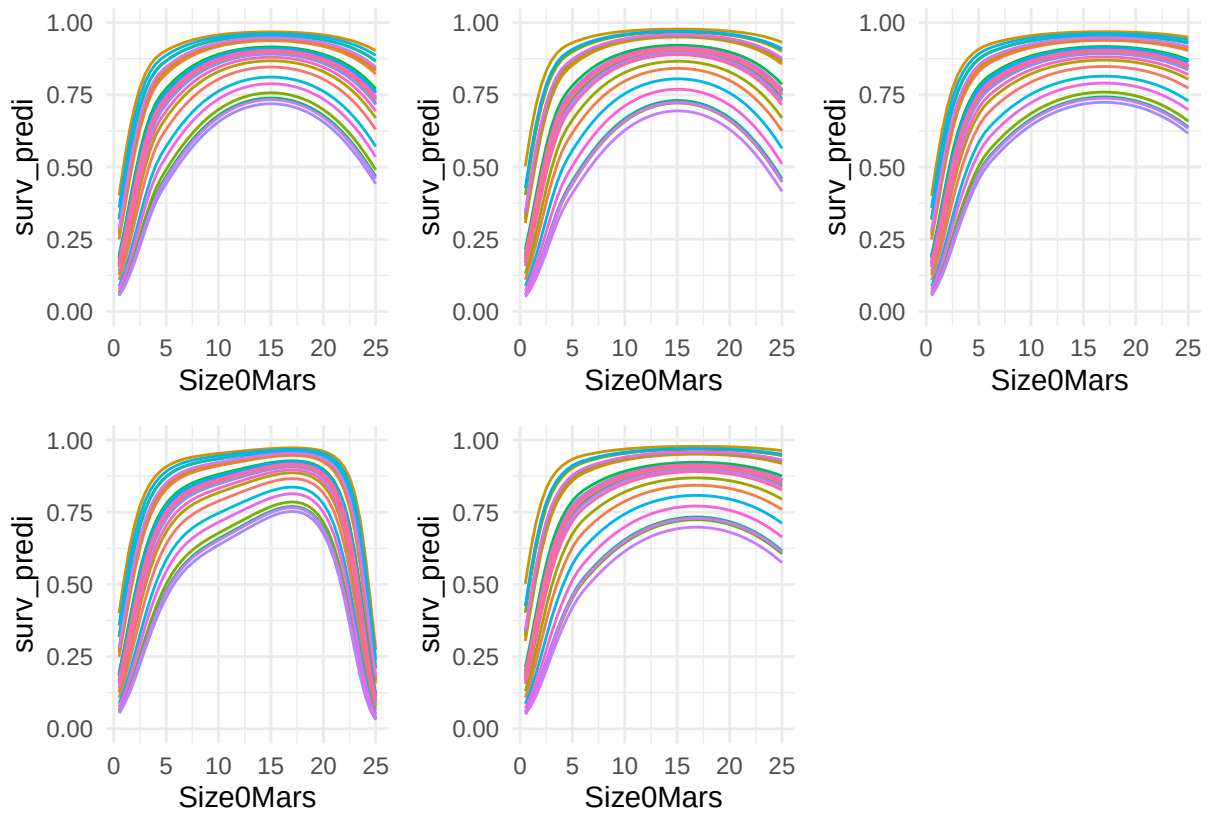


AIC

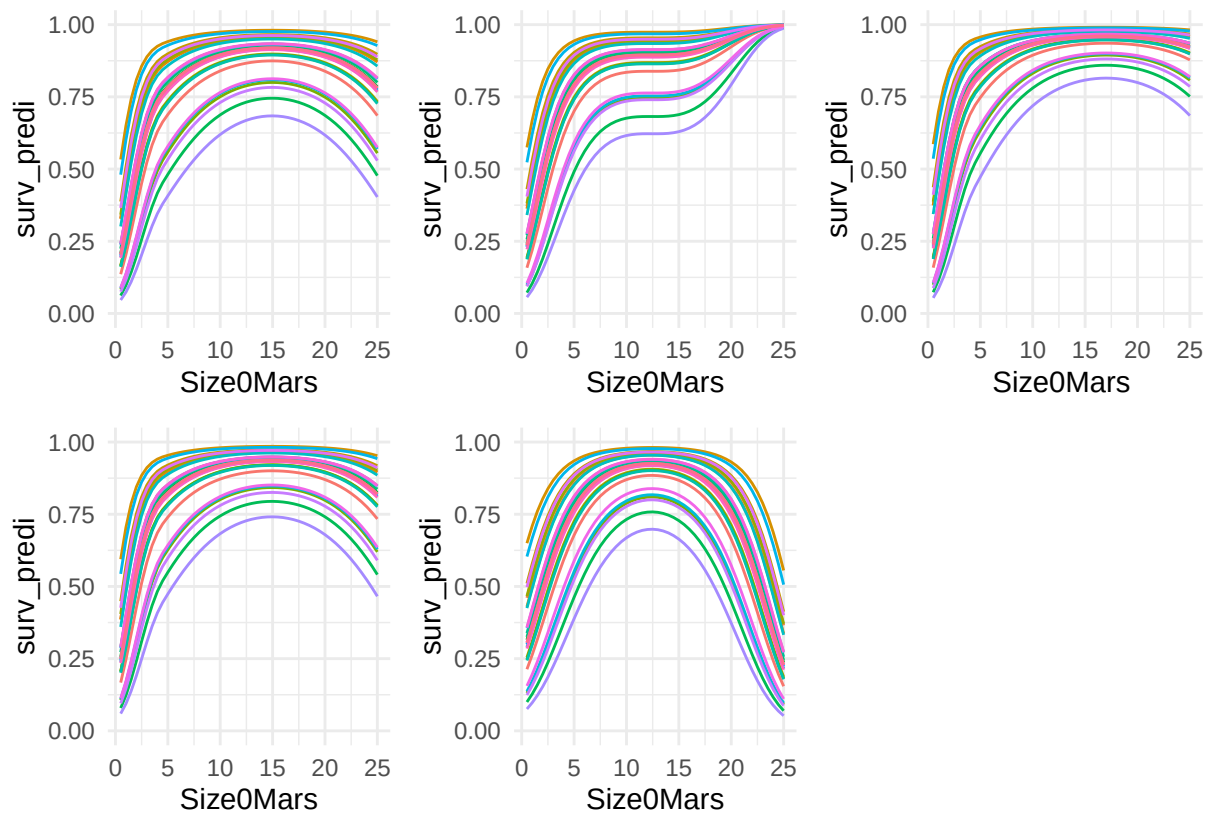


BIC

See year effect (population fixed and age=5)

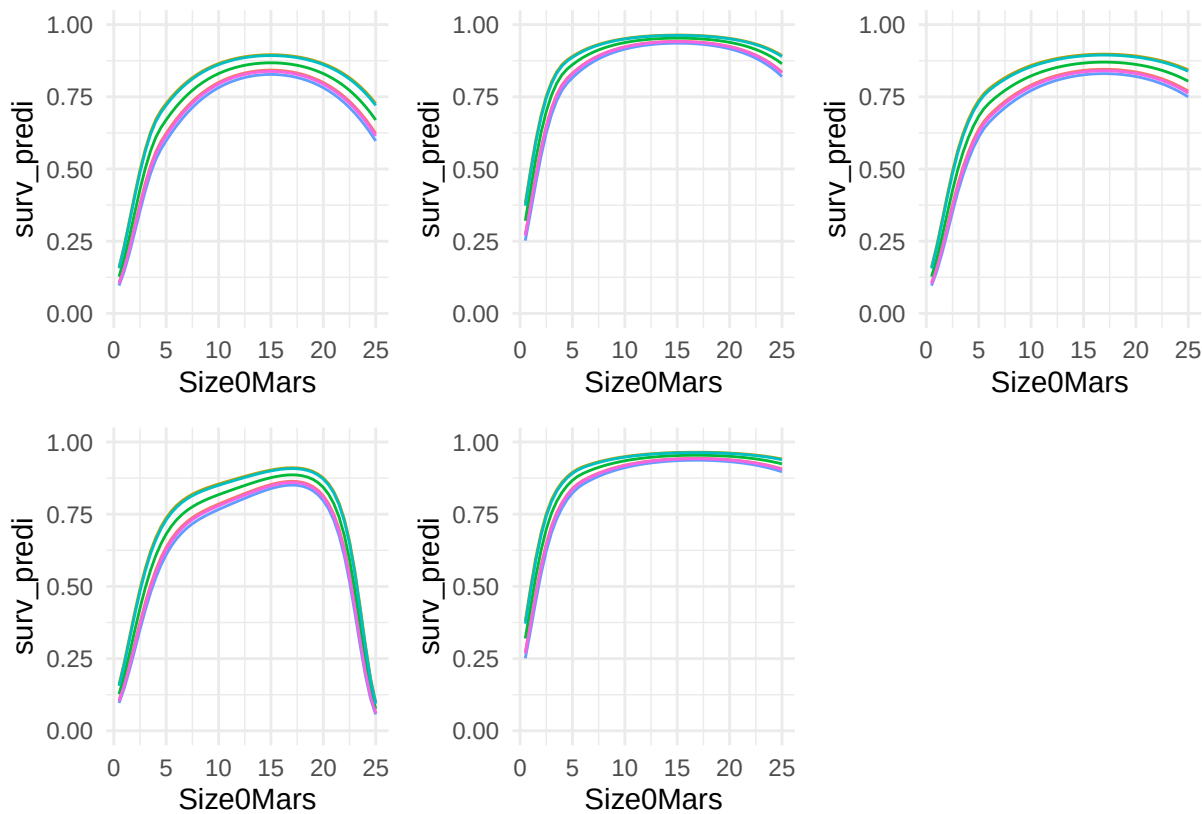


AIC

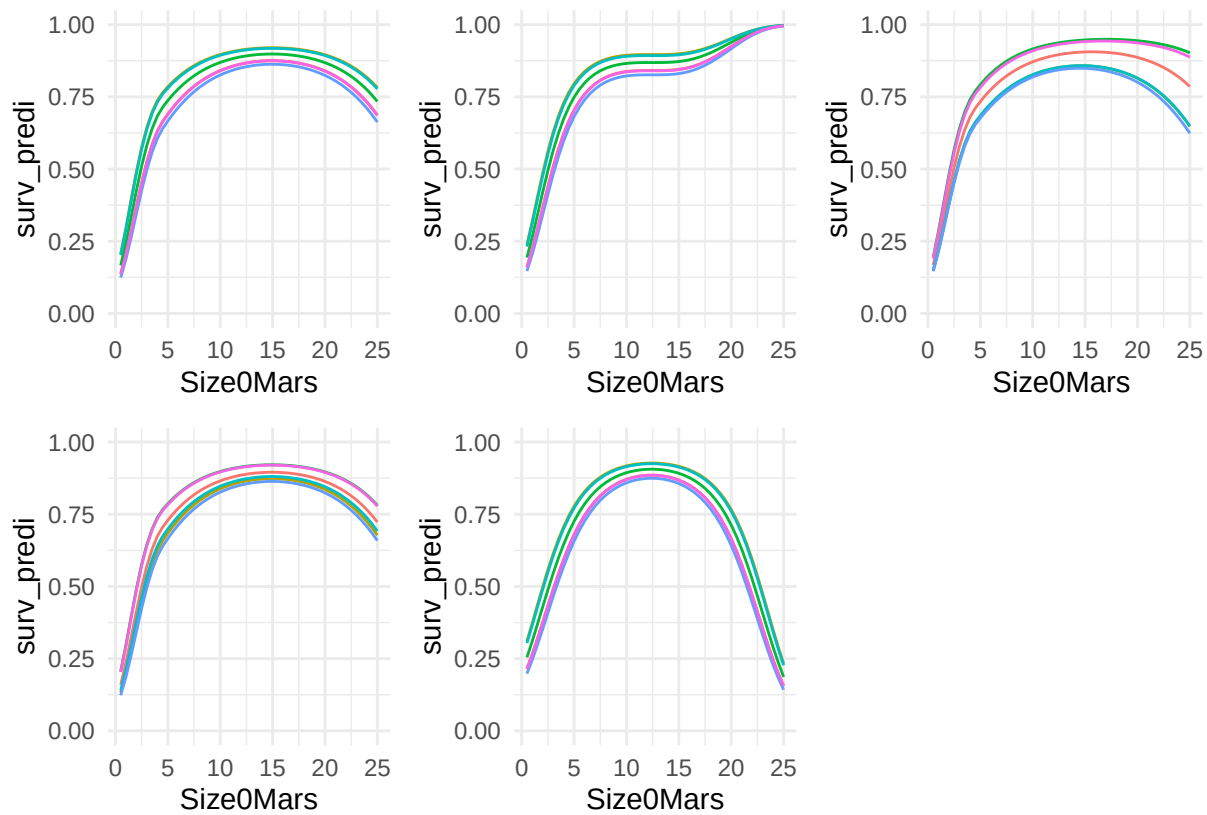


BIC

See population effect (year fixed and age=5)



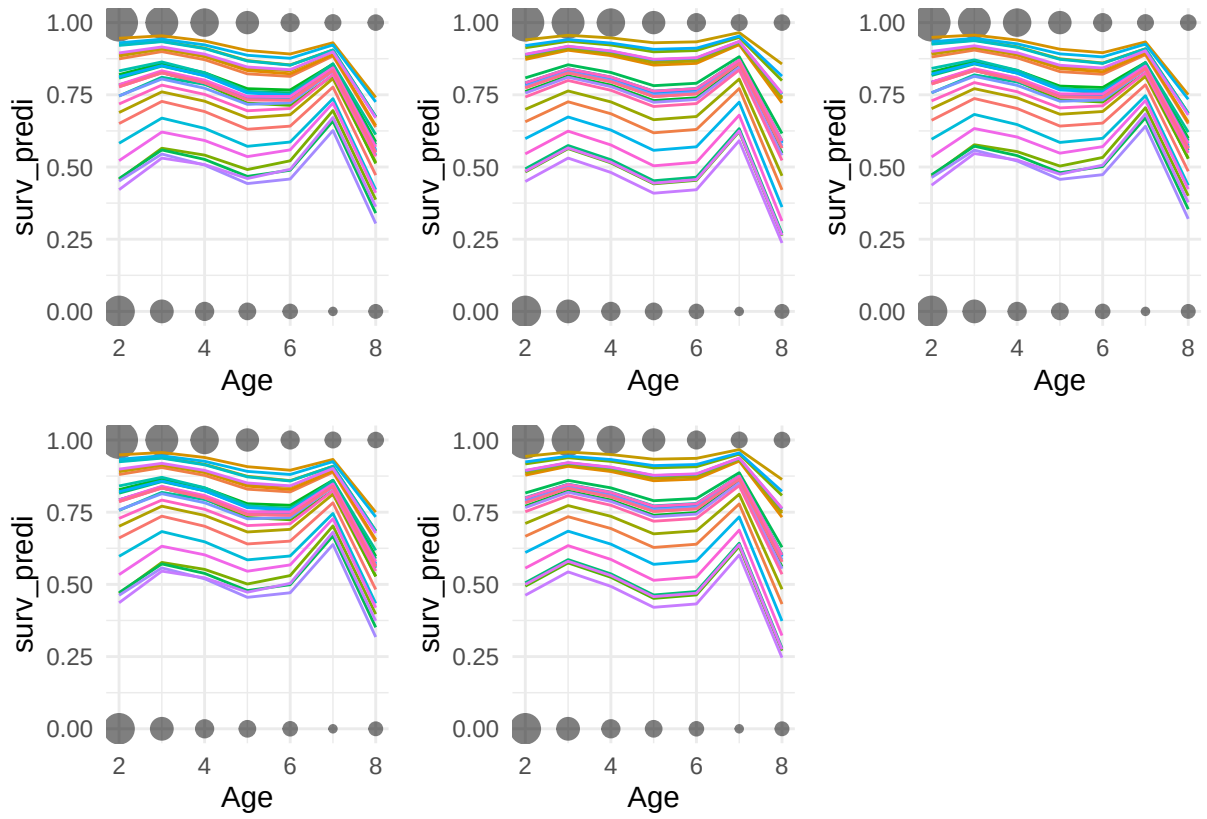
AIC



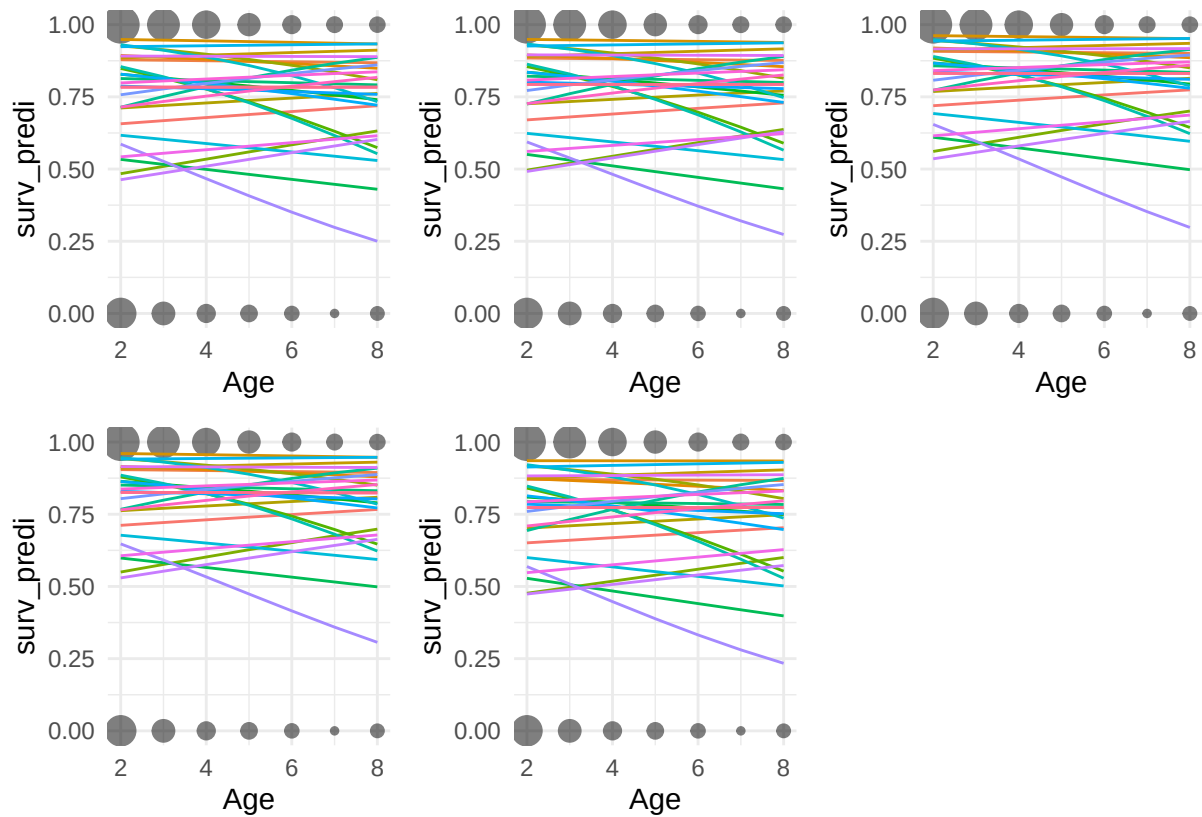
BIC

As a function of age (size=5)

See year effect (population fixed)



AIC



BIC

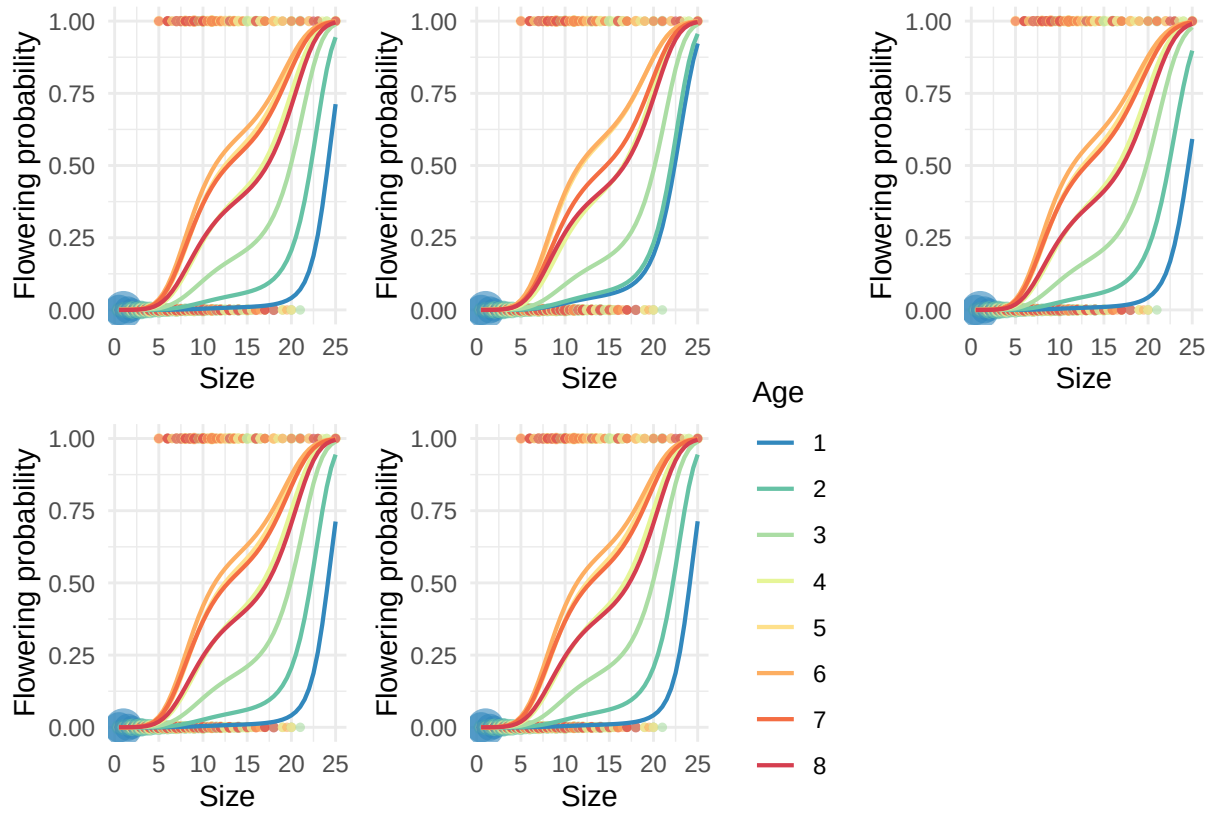
Flowering probability

AIC

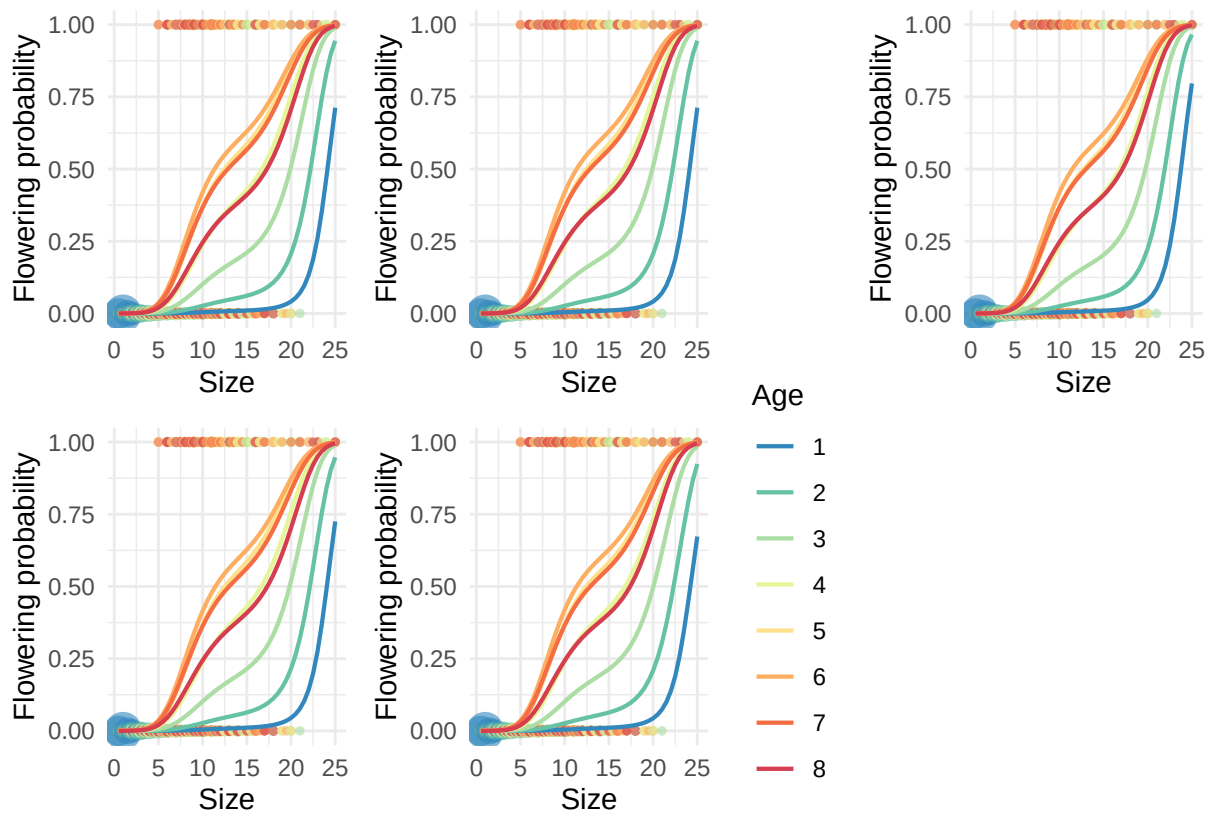
BIC

As a function of size

See effect of age



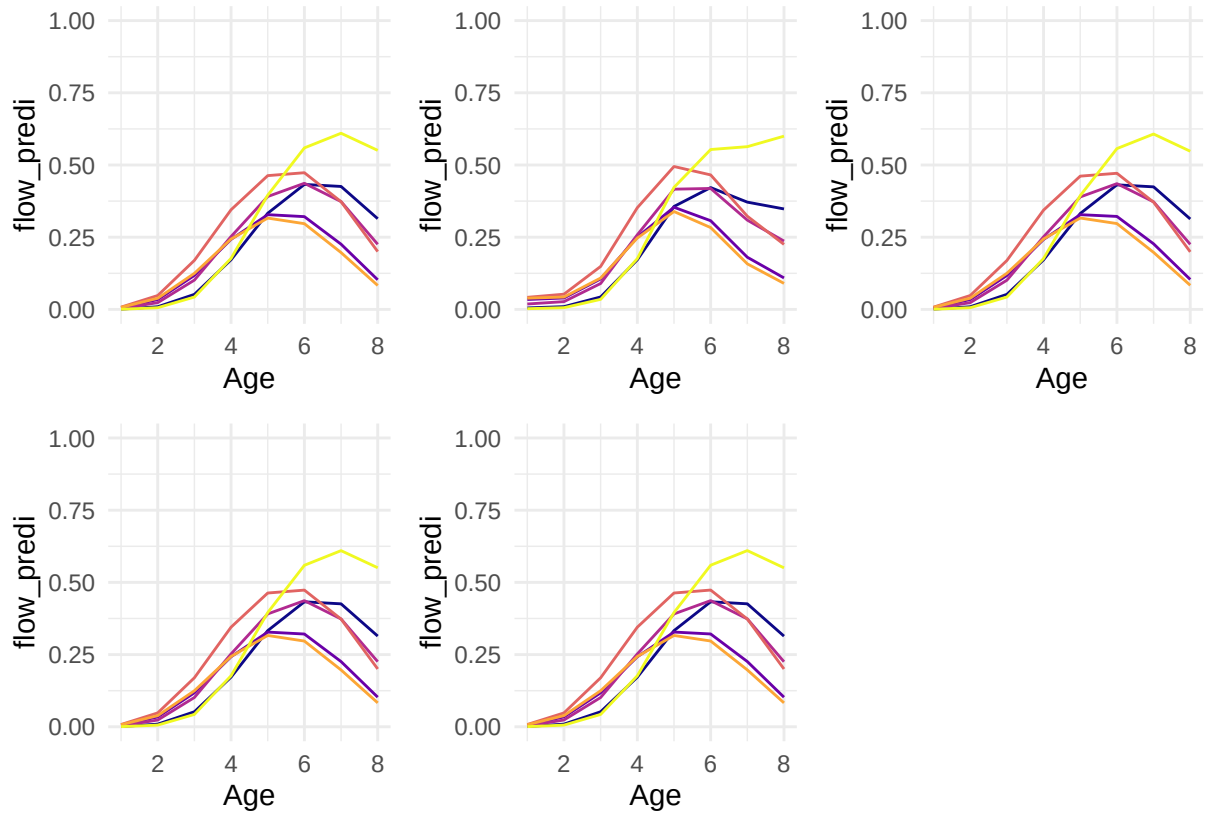
AIC



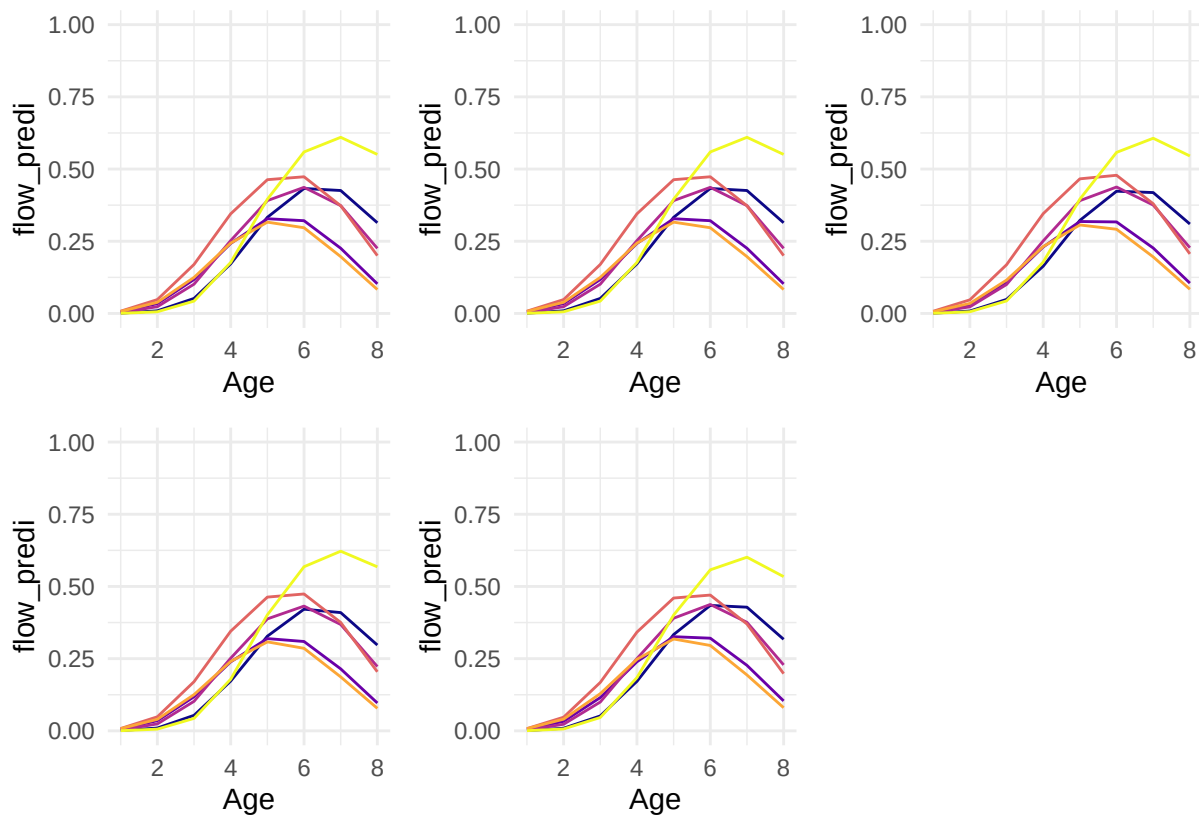
BIC

As a function of age (size fixed)

See population effect (mean over the years)

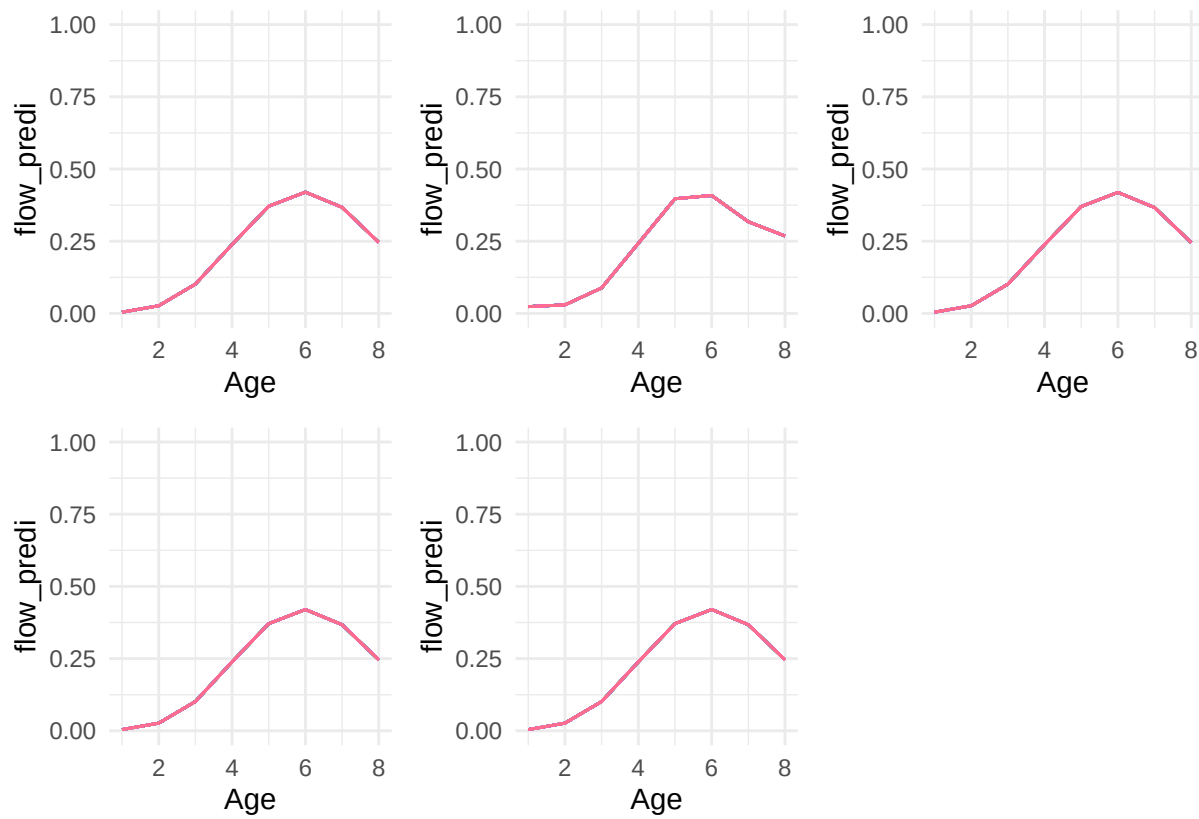


AIC

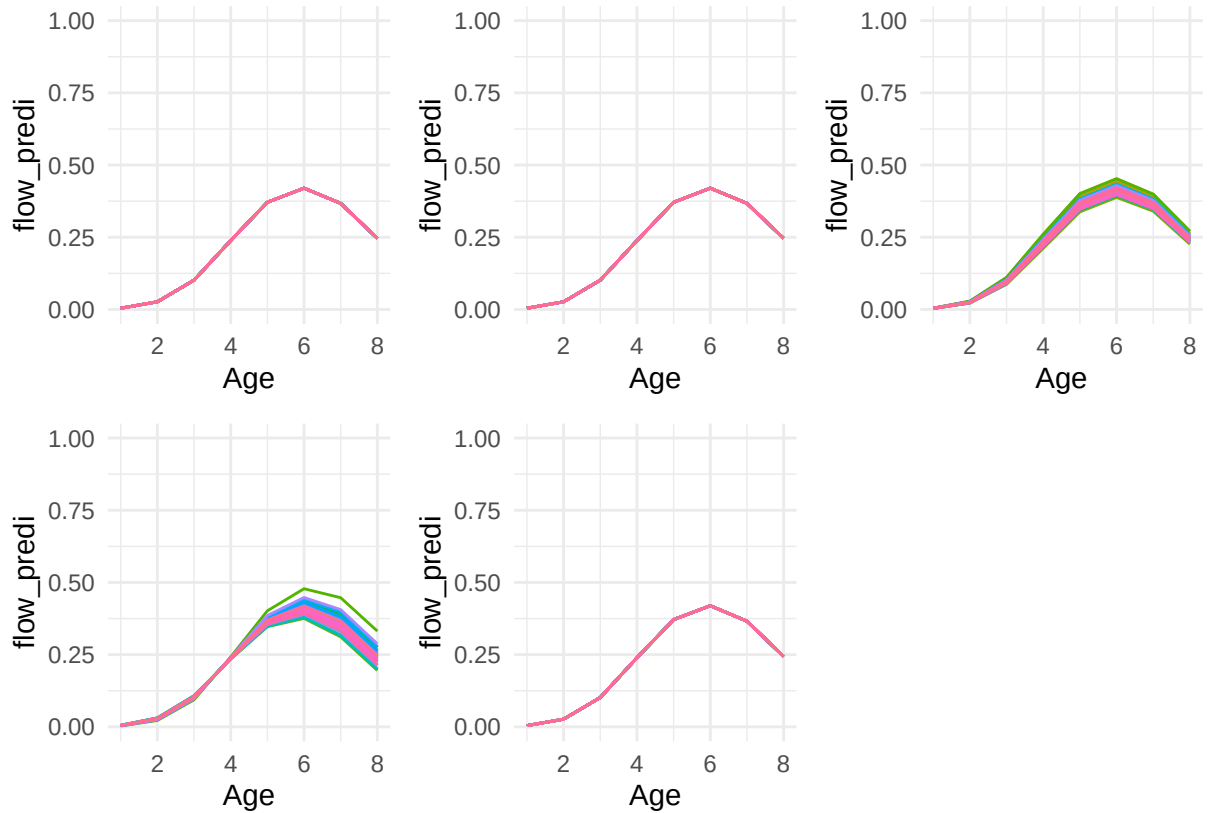


BIC

See year effect (mean over the populations)



AIC



BIC

Growth

AIC

```
AGrowthglm1 <- fitme(Size1Mars ~ 1 + poly(Size0Mars,3) +
  bs(Age,degree=2,knots=6.5) + (Size0Mars+Age|year) +
  (1|Pop),
  resid.model = ~ log(Size0Mars)+log(Age),
  data=growthdata)

AGrowthglm2 <- fitme(Size1Mars ~ 1 + poly(Size0Mars,3) +
  bs(Age,degree=2,knots=6.5) + (Size0Mars+Age|year) +
  (Size0Mars|Pop),
  resid.model = ~ log(Size0Mars)+log(Age),
  data=growthdata)

AGrowthglm3 <- fitme(Size1Mars ~ 1 + poly(Size0Mars,3) +
  bs(Age,degree=2,knots=6.5) + (Size0Mars+Age|year) +
  (Size0Mars+Age|Pop),
  resid.model = ~ log(Size0Mars)+log(Age),
  data=growthdata)

AGrowthglm4 <- fitme(Size1Mars ~ 1 + bs(Size0Mars,df=5,degree=3) +
  bs(Age,degree=2,knots=6.5) + (Size0Mars+Age|year) +
```

```

      (1|Pop),
      resid.model = ~ log(Size0Mars)+log(Age),
      data=growthdata)

AGrowthglm5 <- fitme(Size1Mars ~ 1 + bs(Size0Mars,df=5,degree=3) +
      bs(Age,degree=2,knots=6.5) + (Size0Mars+Age|year) +
      (Size0Mars|Pop),
      resid.model = ~ log(Size0Mars)+log(Age),
      data=growthdata)

```

BIC

```

BGrowthglm1 <- fitme(Size1Mars ~ 1 + poly(Size0Mars,3) + poly(Age,2) +
      (Size0Mars+Age|year) + (1|Pop),
      resid.model = ~ log(Size0Mars)+log(Age),
      data=growthdata)

BGrowthglm2 <- fitme(Size1Mars ~ 1 + poly(Size0Mars,3) + poly(Age,2) +
      (Size0Mars+Age|year) + (Size0Mars+Age|Pop),
      resid.model = ~ log(Size0Mars)+log(Age),
      data=growthdata)

BGrowthglm3 <- fitme(Size1Mars ~ 1 + bs(Size0Mars,df=3,degree=2) + poly(Age,2) +
      (Size0Mars+Age|year) + (1|Pop),
      resid.model = ~ log(Size0Mars)+log(Age),
      data=growthdata)

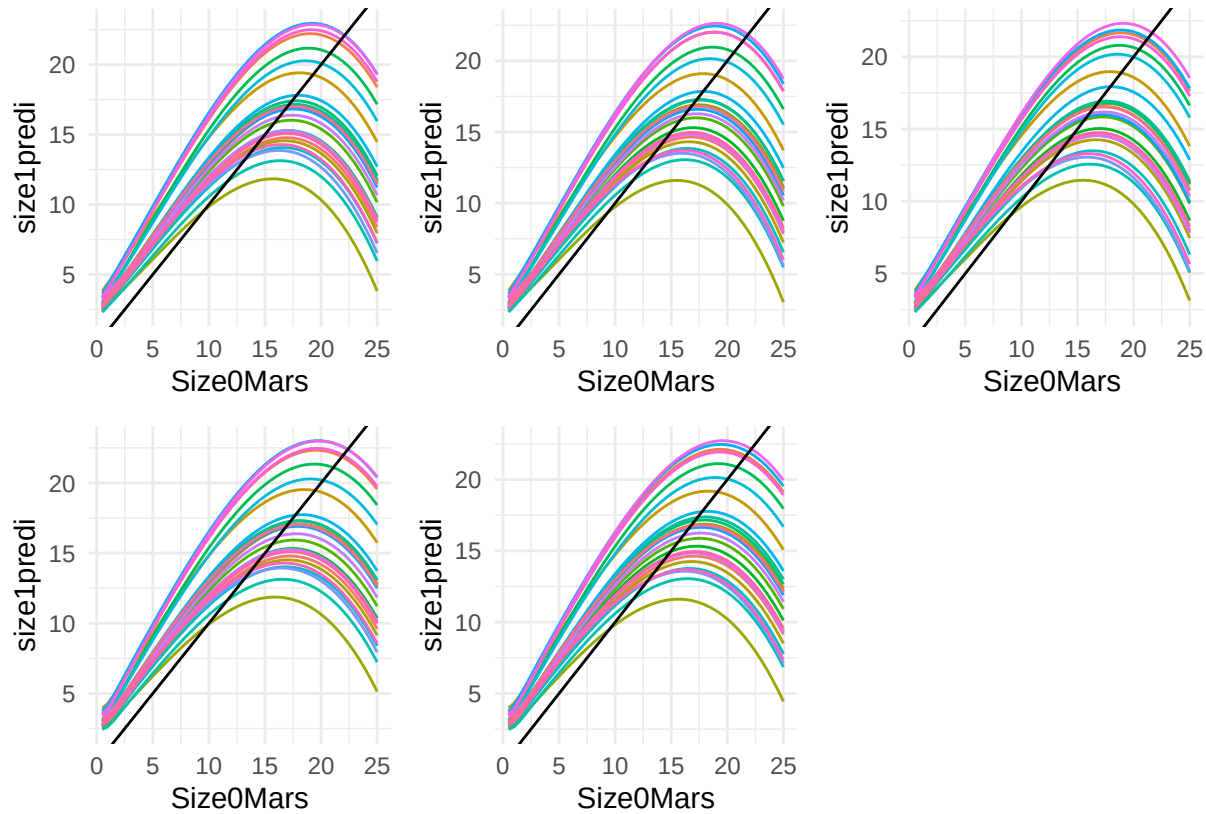
BGrowthglm4 <- fitme(Size1Mars ~ 1 + bs(Size0Mars,df=3,degree=2) + poly(Age,2) +
      (Size0Mars+Age|year) + (Size0Mars|Pop),
      resid.model = ~ log(Size0Mars)+log(Age),
      data=growthdata)

BGrowthglm5 <- fitme(Size1Mars ~ 1 + bs(Size0Mars,df=3,degree=2) + poly(Age,2) +
      (Size0Mars+Age|year) + (Size0Mars+Age|Pop),
      resid.model = ~ log(Size0Mars)+log(Age),
      data=growthdata)

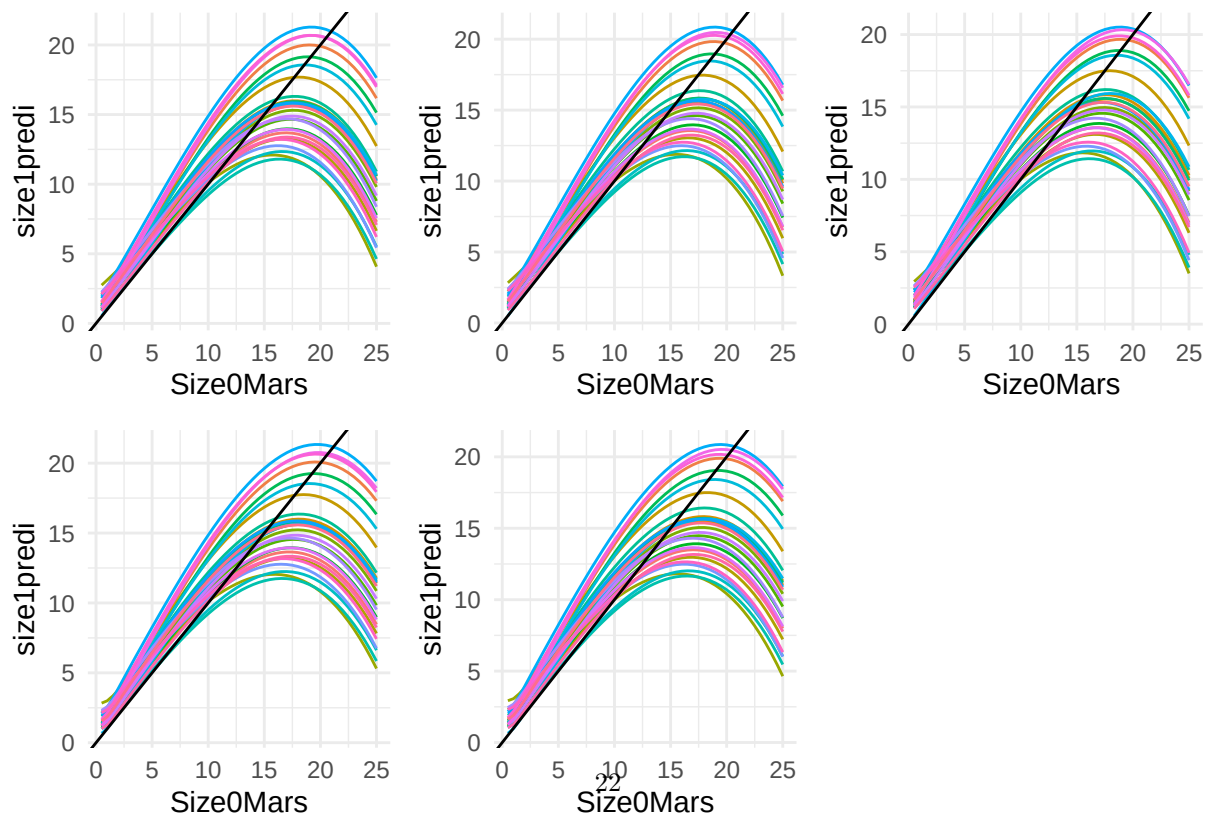
```

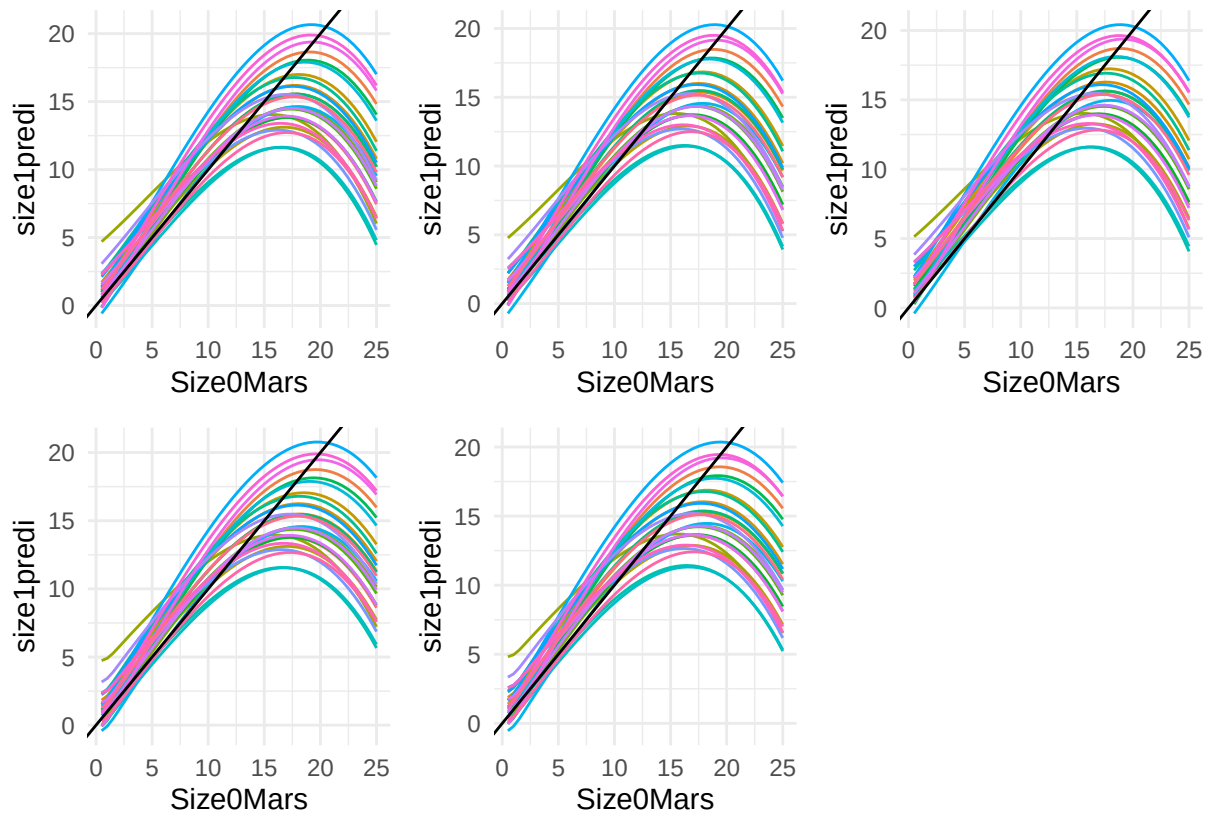
Size at $t+1$ as a function of size at t (age 1, 4, 8)

See year effect (population fixed)

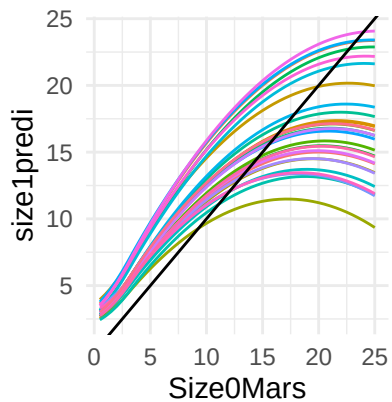
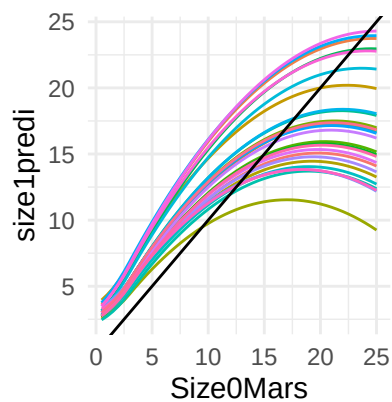
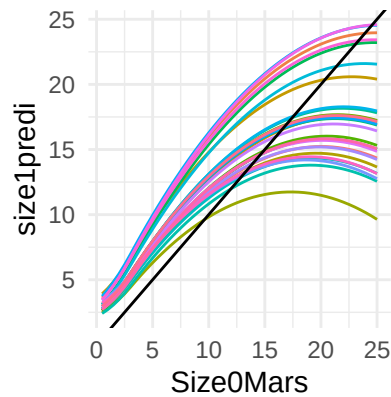
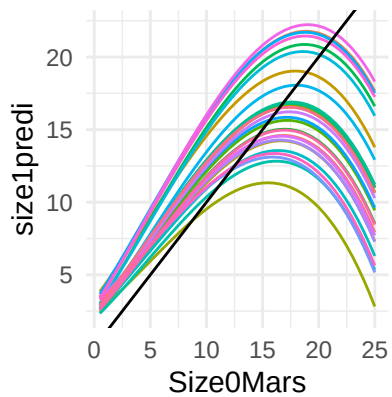
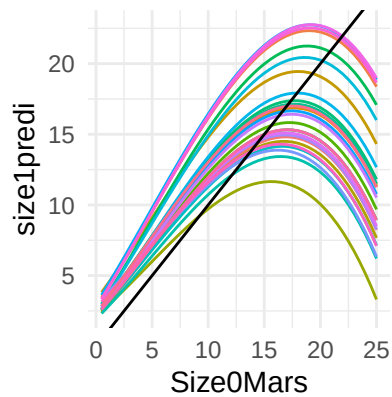


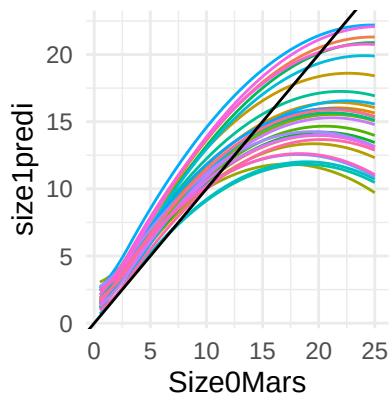
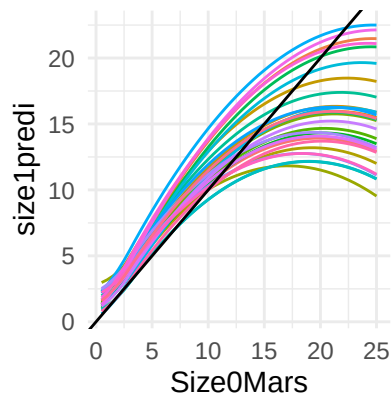
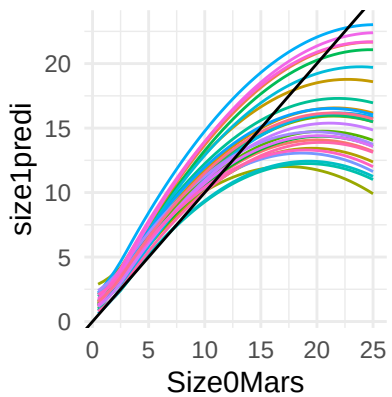
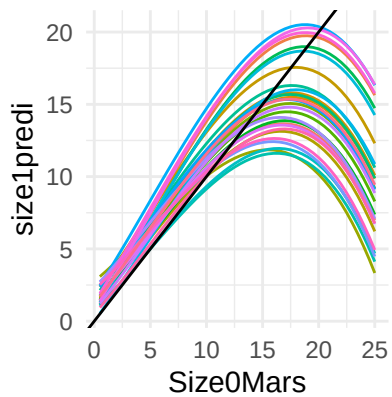
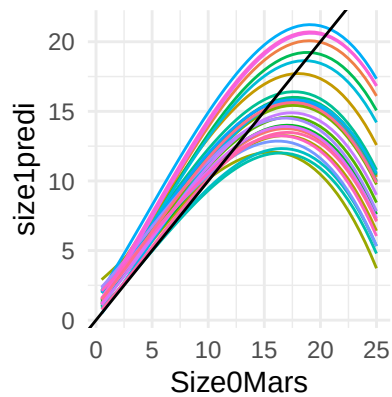
AIC age 1 - 4 - 8

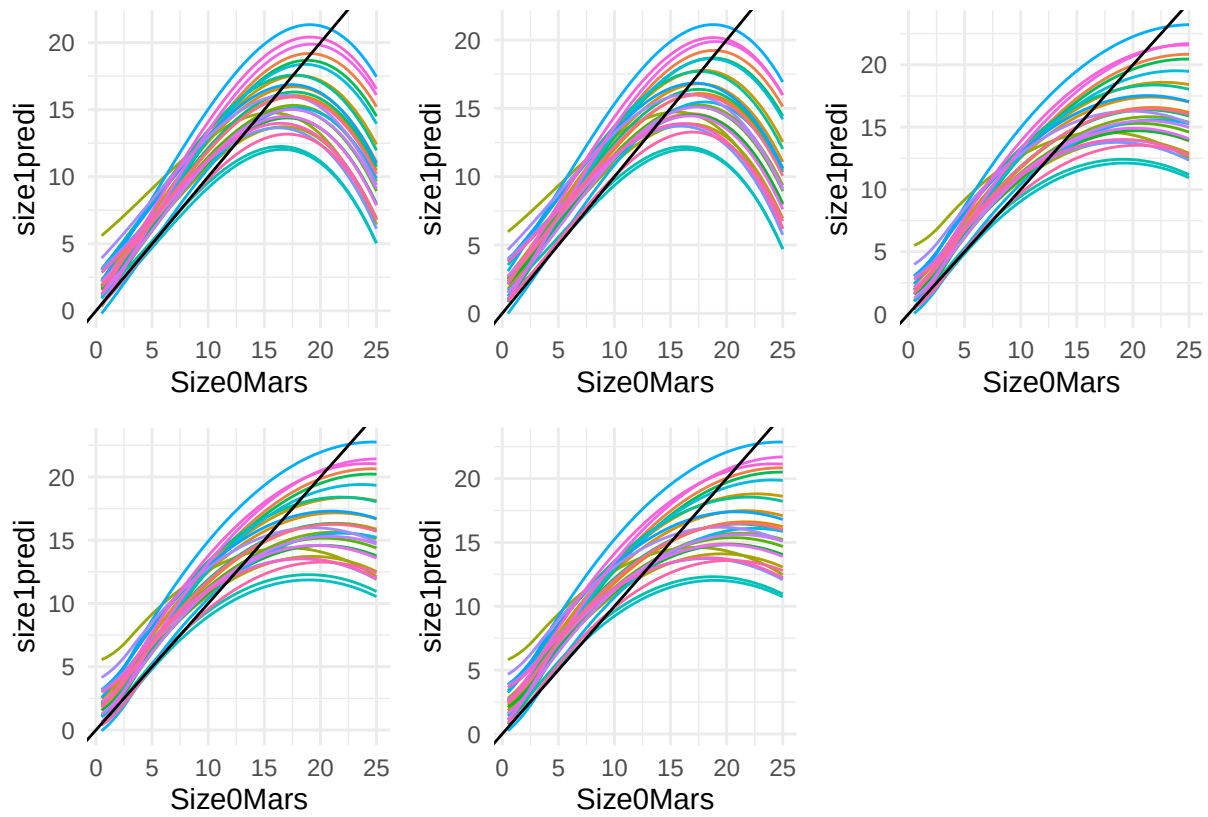




BIC age 1 - 4 - 8

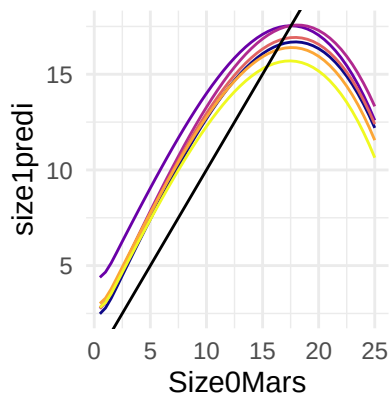
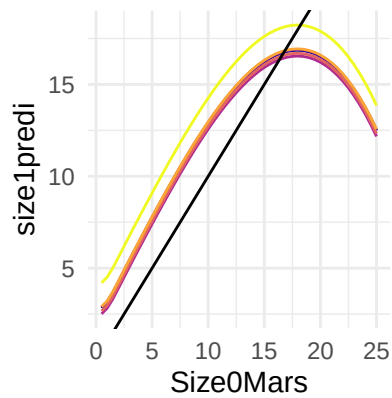
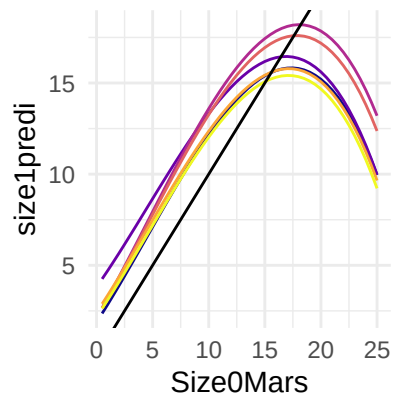
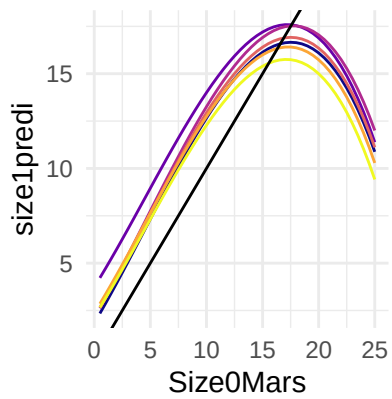
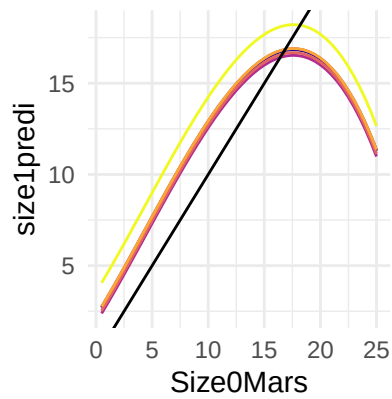


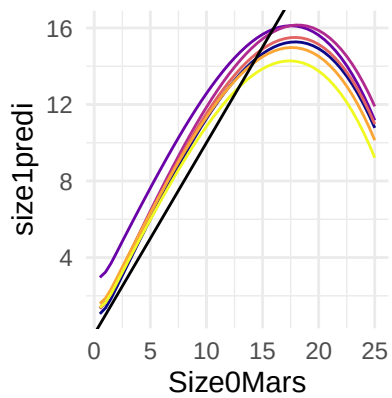
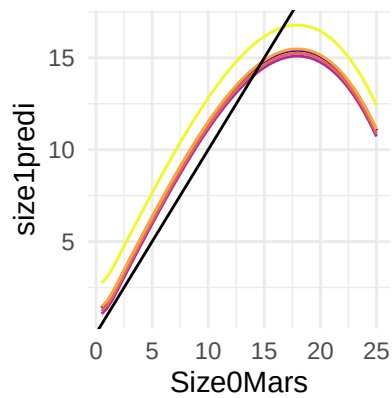
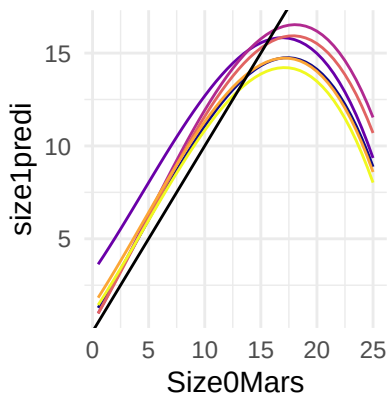
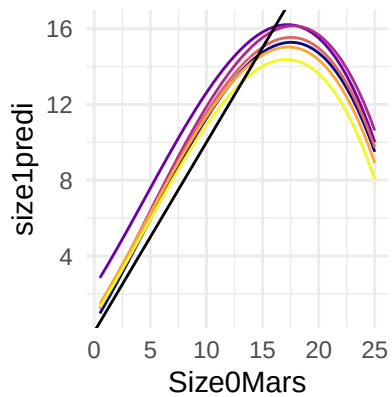
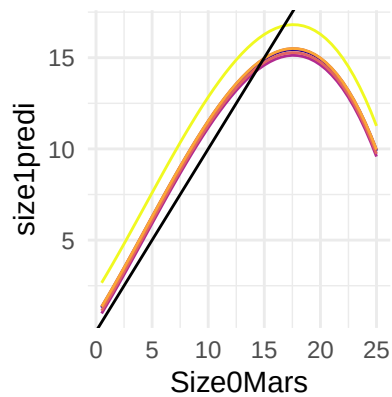


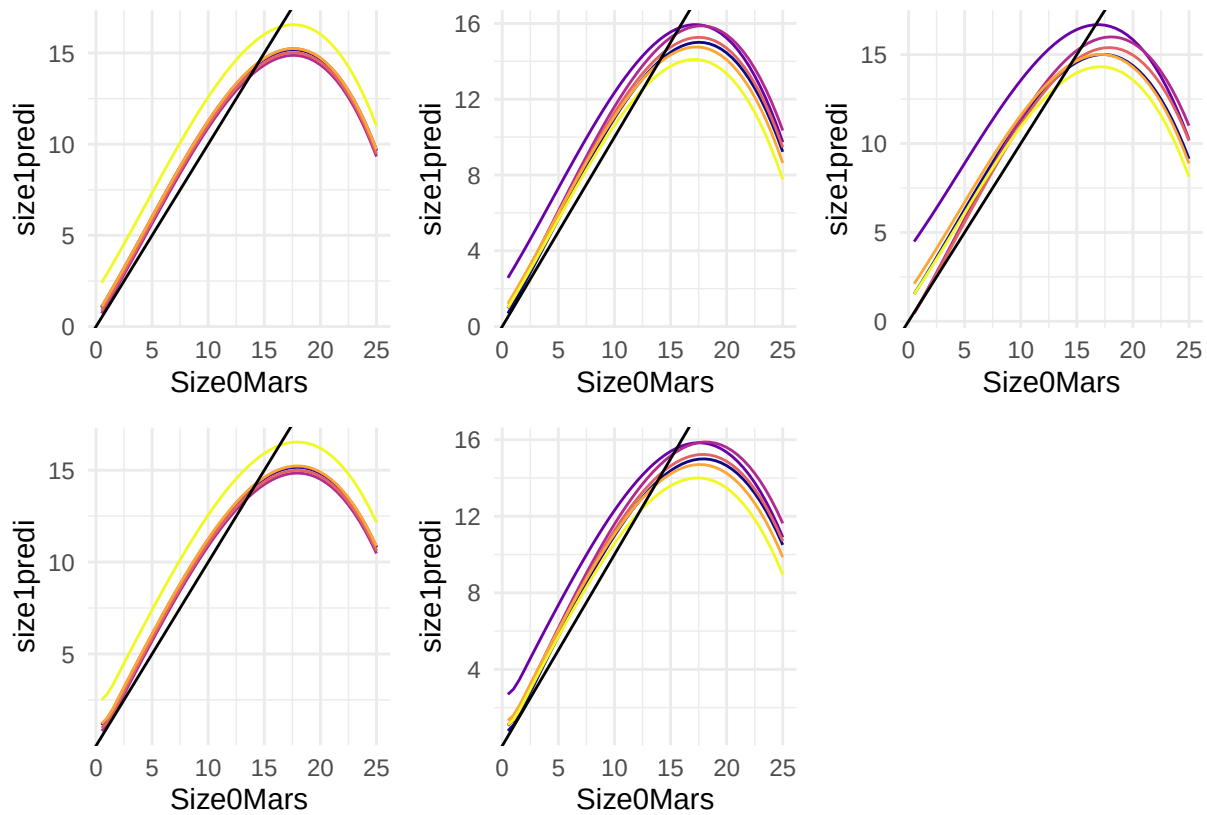


See population effect (year fixed)

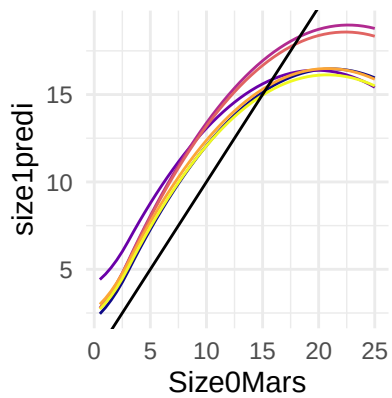
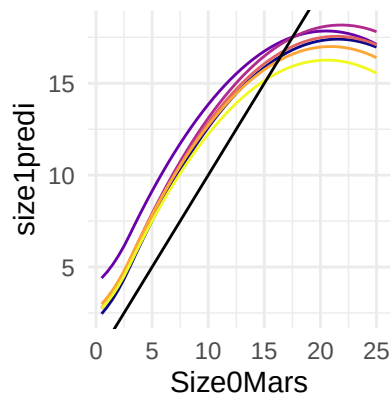
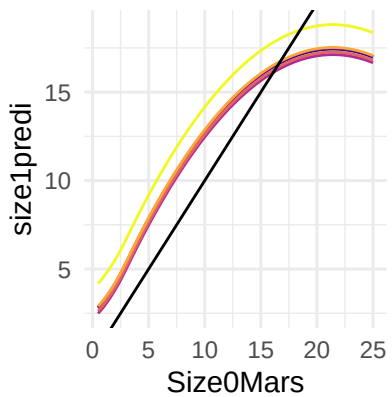
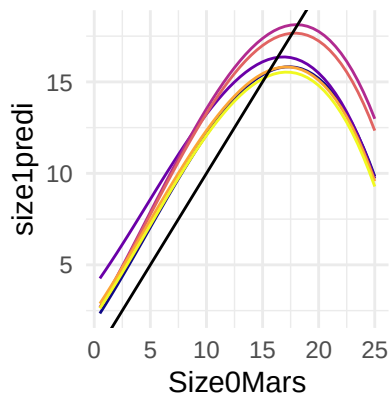
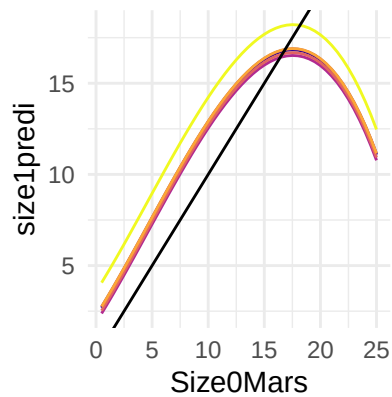
AIC age 1 - 4 - 8

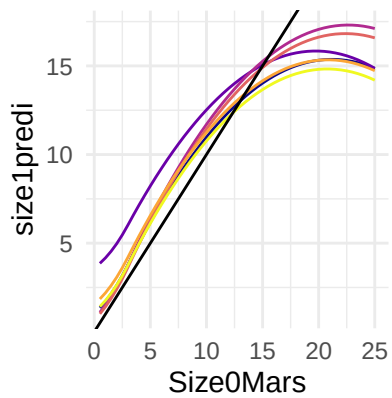
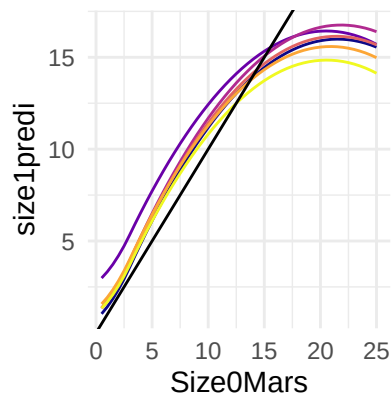
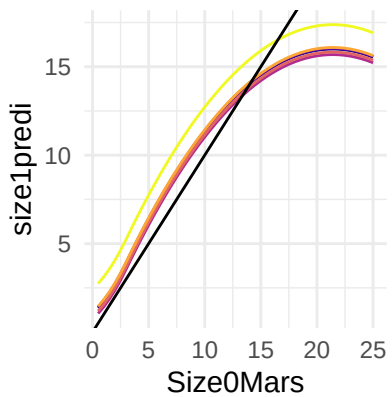
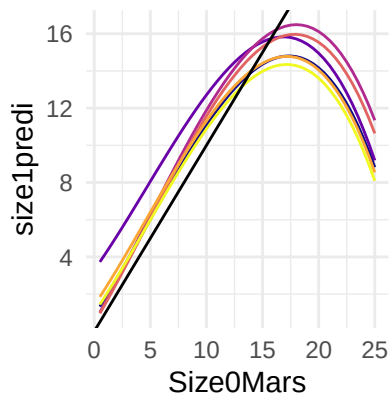
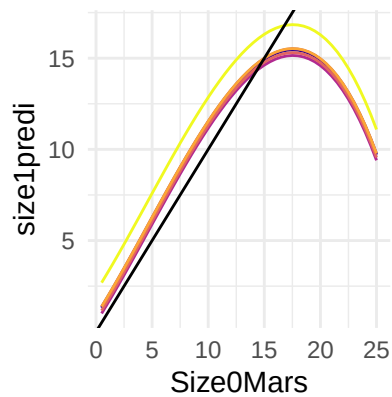


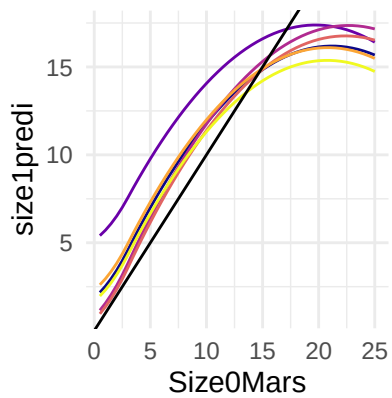
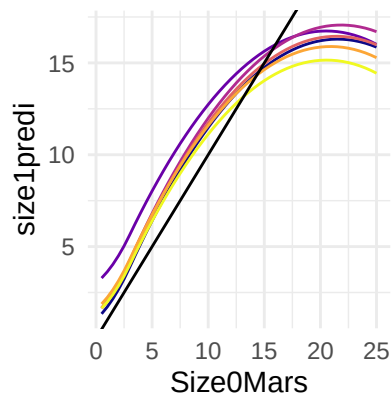
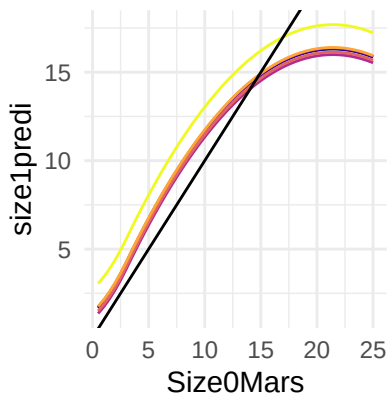
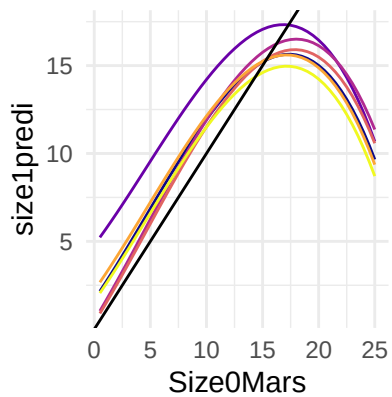
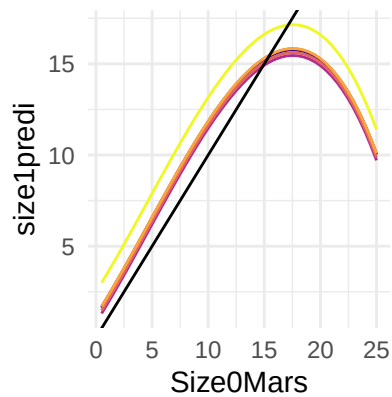




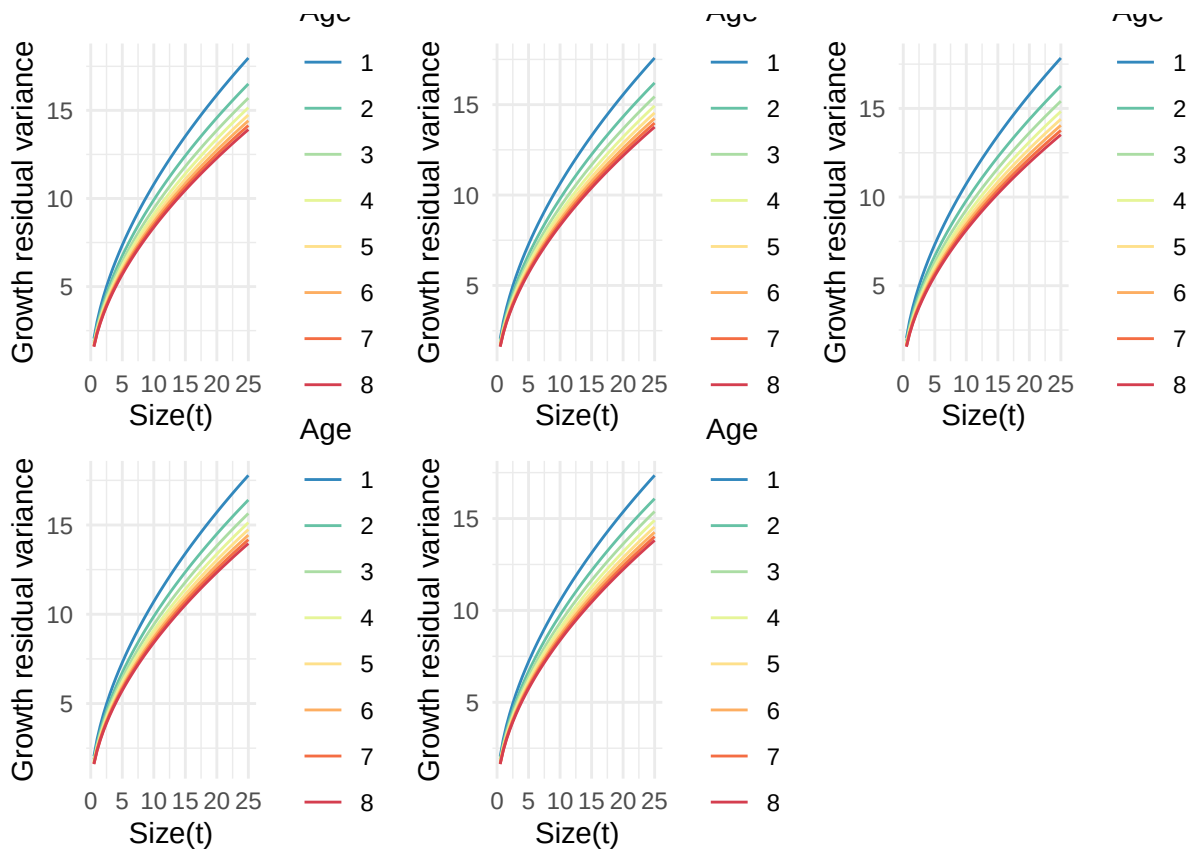
BIC age 1 - 4 - 8



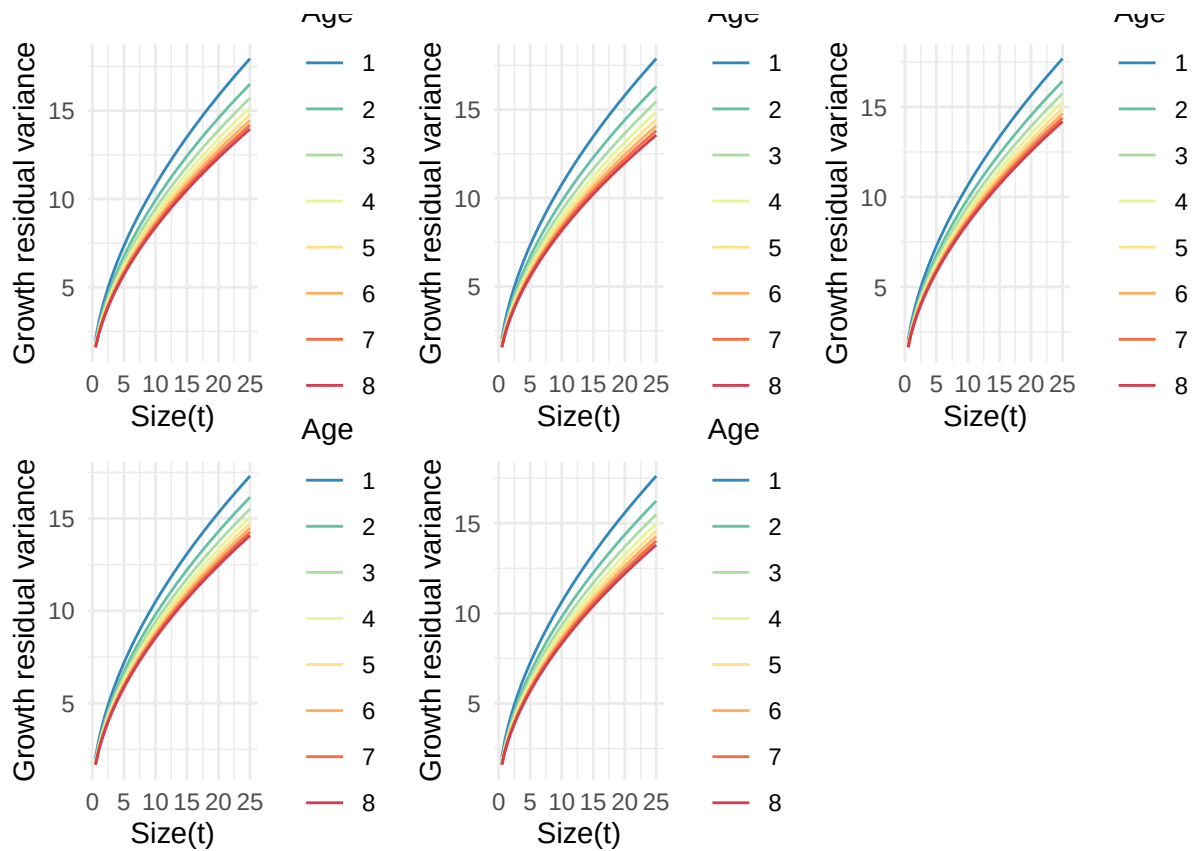




Residual variance



AIC



BIC

Number of capitula (fecundity)

AIC

```
ACptlglm1 <- fitme(log(Capitule) ~ 1 + Size0Mars + (Age|year),
  data=cptldata)

ACptlglm2 <- fitme(log(Capitule) ~ 1 + Size0Mars + (Age|year) + (1|Pop),
  data=cptldata)

ACptlglm3 <- fitme(log(Capitule) ~ 1 + poly(Size0Mars,2) + (Age|year),
  data=cptldata)

ACptlglm4 <- fitme(log(Capitule) ~ 1 + Size0Mars + Age + (Age|year),
  data=cptldata)

ACptlglm5 <- fitme(log(Capitule) ~ 1 + Size0Mars + (1|year),
  data=cptldata)
```

BIC

```
BCptlglm1 <- fitme(log(Capitule) ~ 1 + Size0Mars + (Age|year),
  data=cptldata)
```

```
BCptlglm2 <- fitme(log(Capitule) ~ 1 + Size0Mars + (Age|year) + (1|Pop),
  data=cptldata)

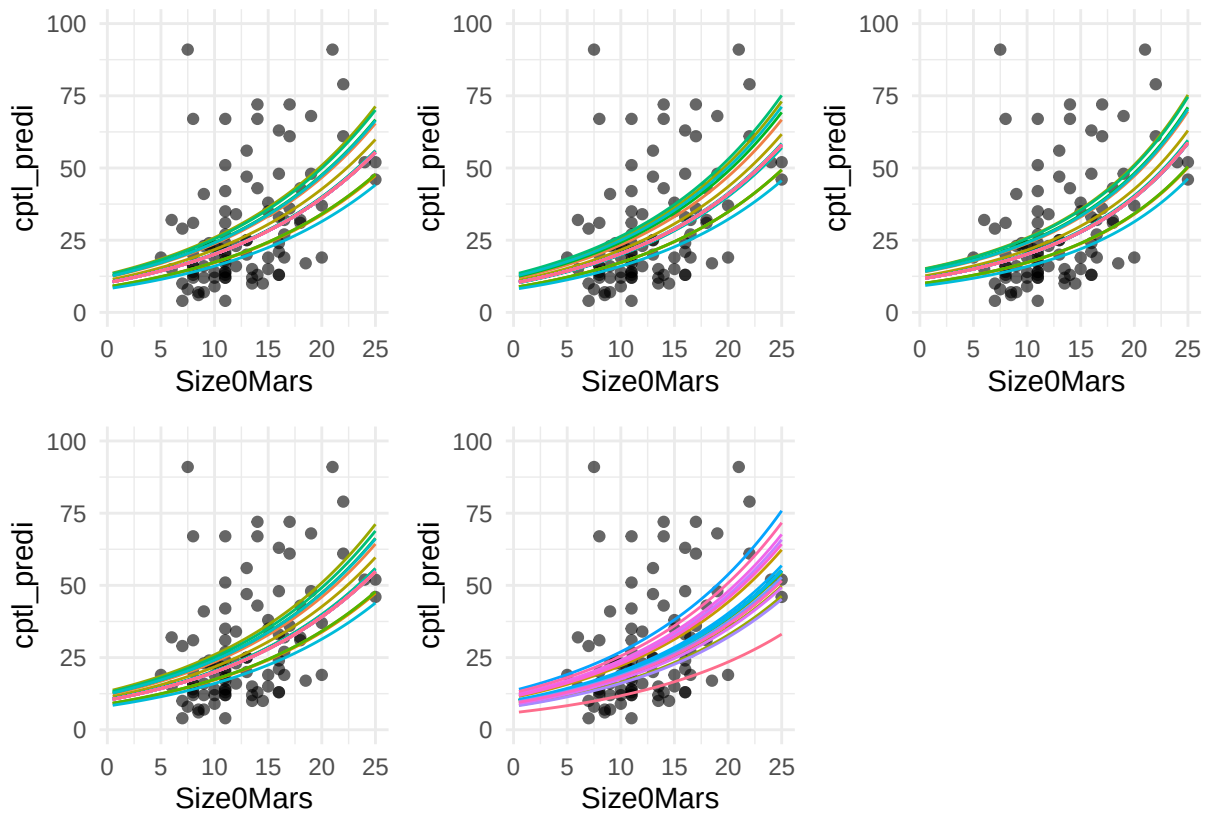
BCptlglm3 <- fitme(log(Capitule) ~ 1 + Size0Mars + (1|year),
  data=cptldata)

BCptlglm4 <- fitme(log(Capitule) ~ 1 + Size0Mars,
  data=cptldata)

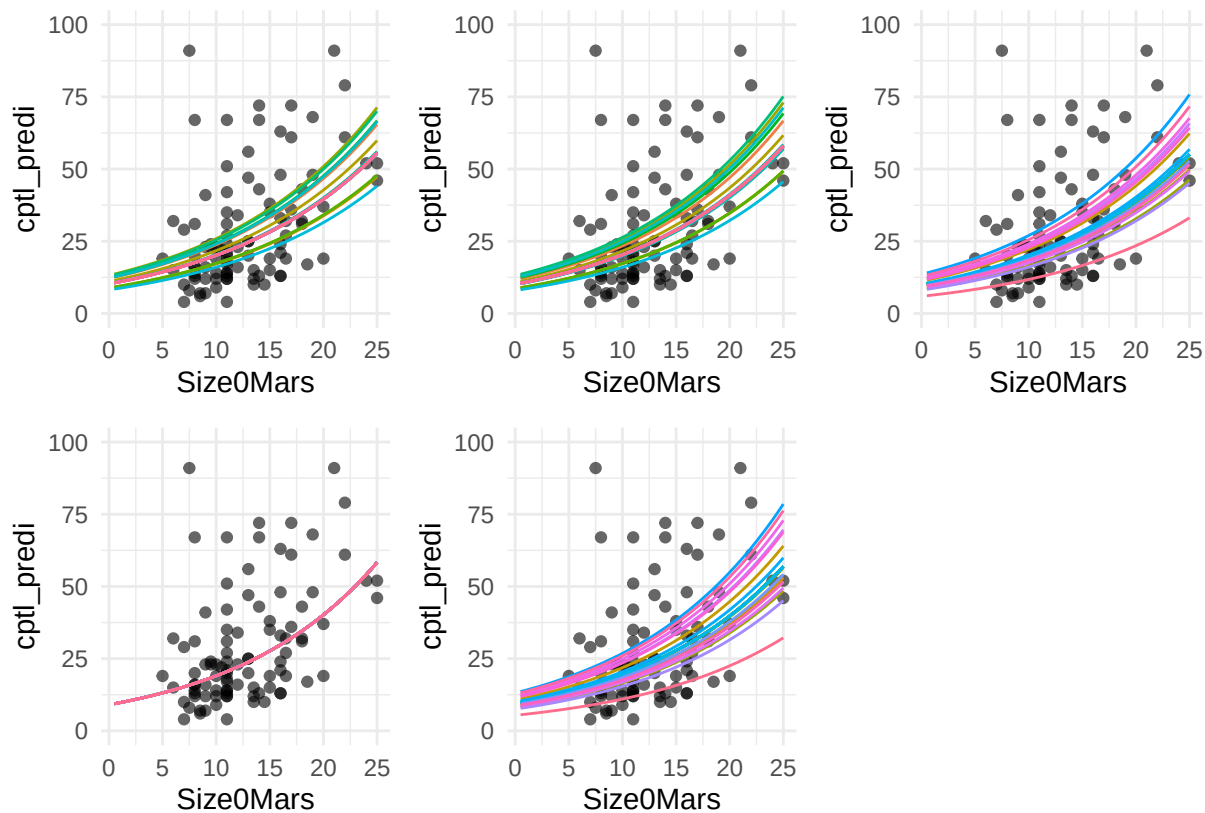
BCptlglm5 <- fitme(log(Capitule) ~ 1 + Size0Mars + (1|year) + (1|Pop),
  data=cptldata)
```

As a function of size

See year effect

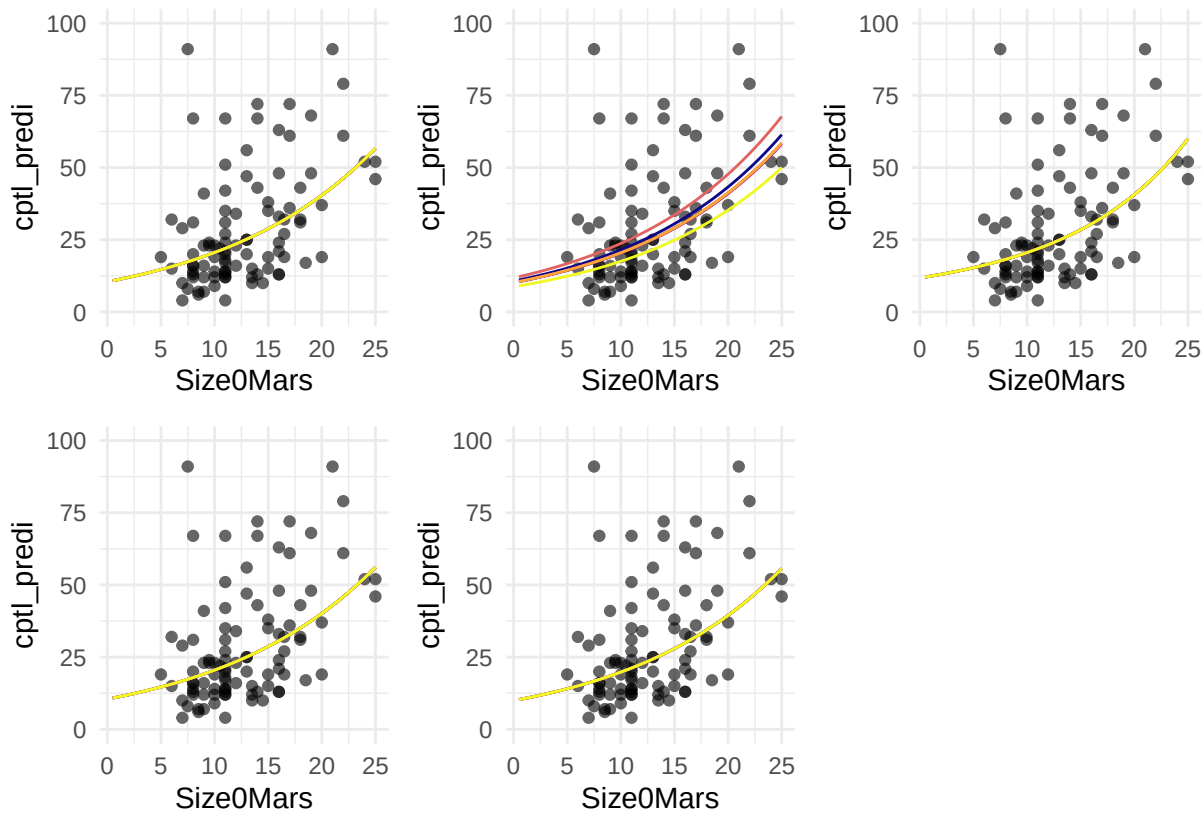


AIC

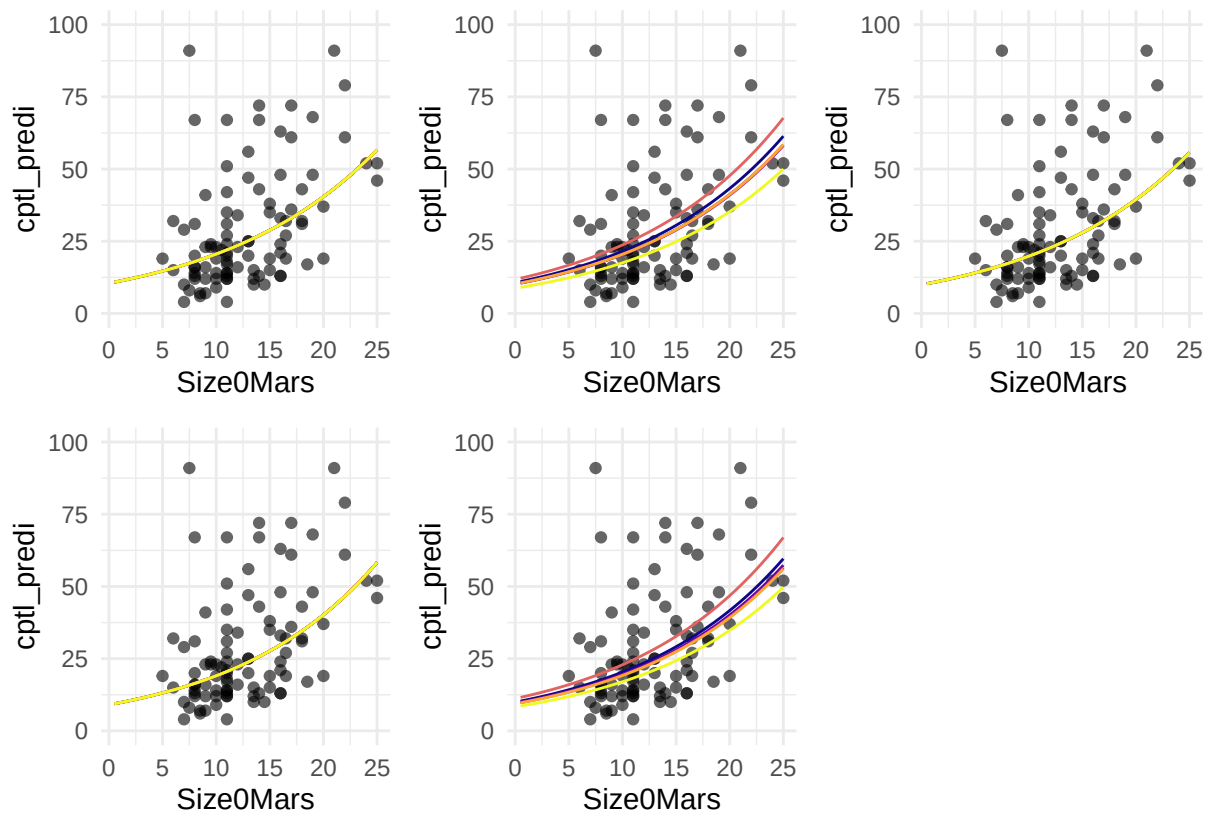


BIC

See population effect



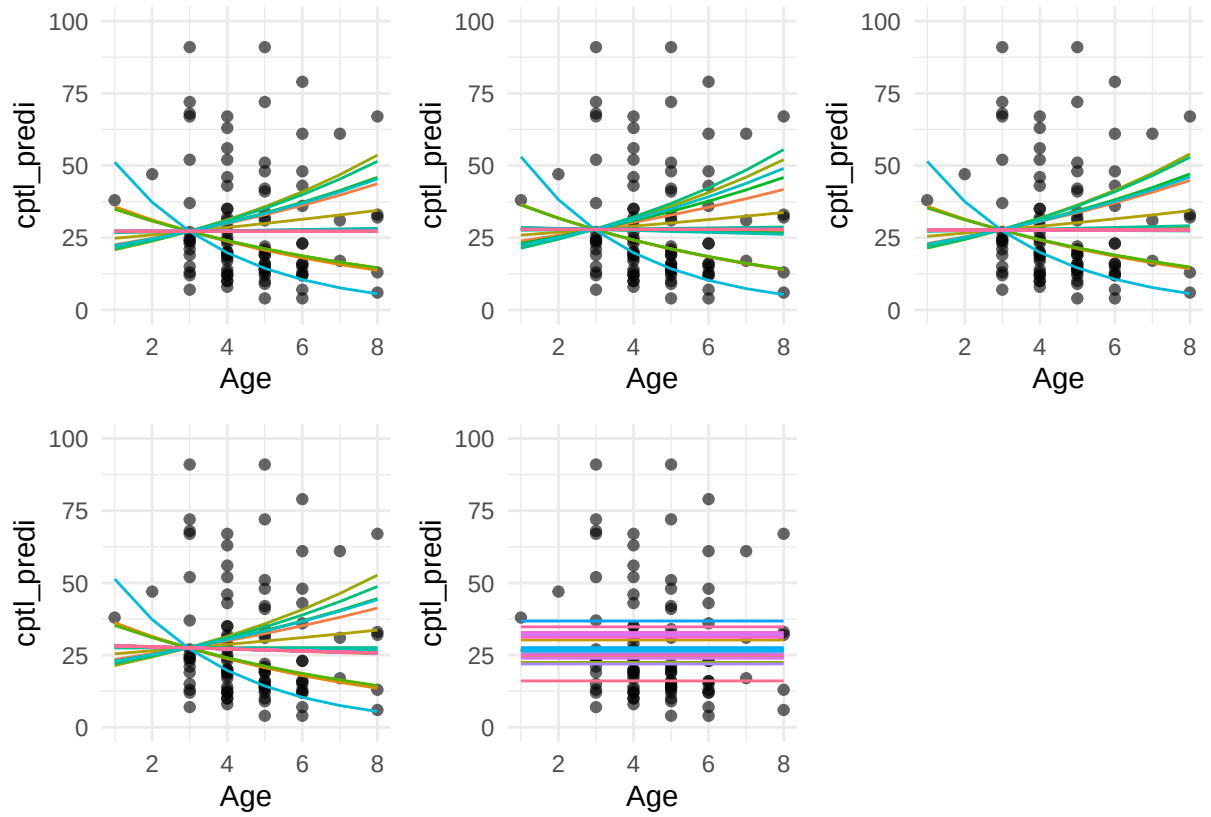
AIC



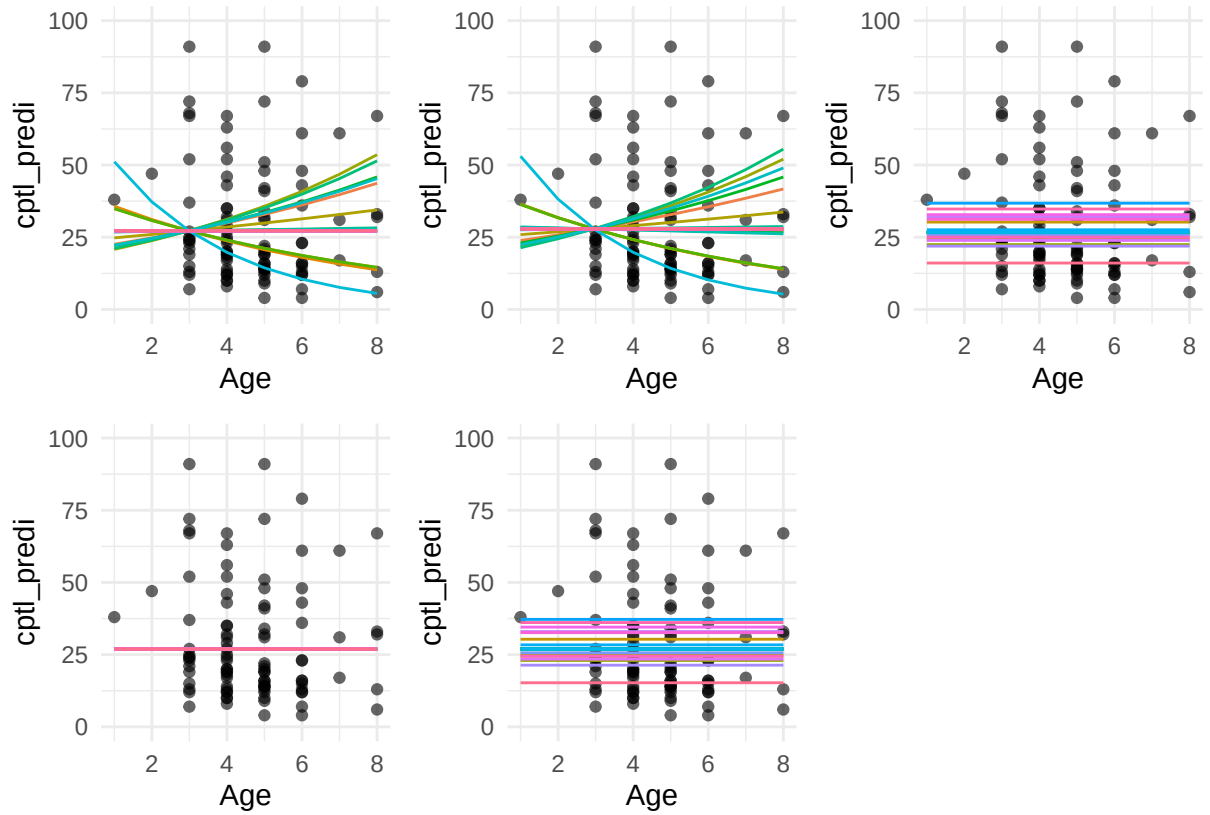
BIC

As a function of age

See year effect



AIC

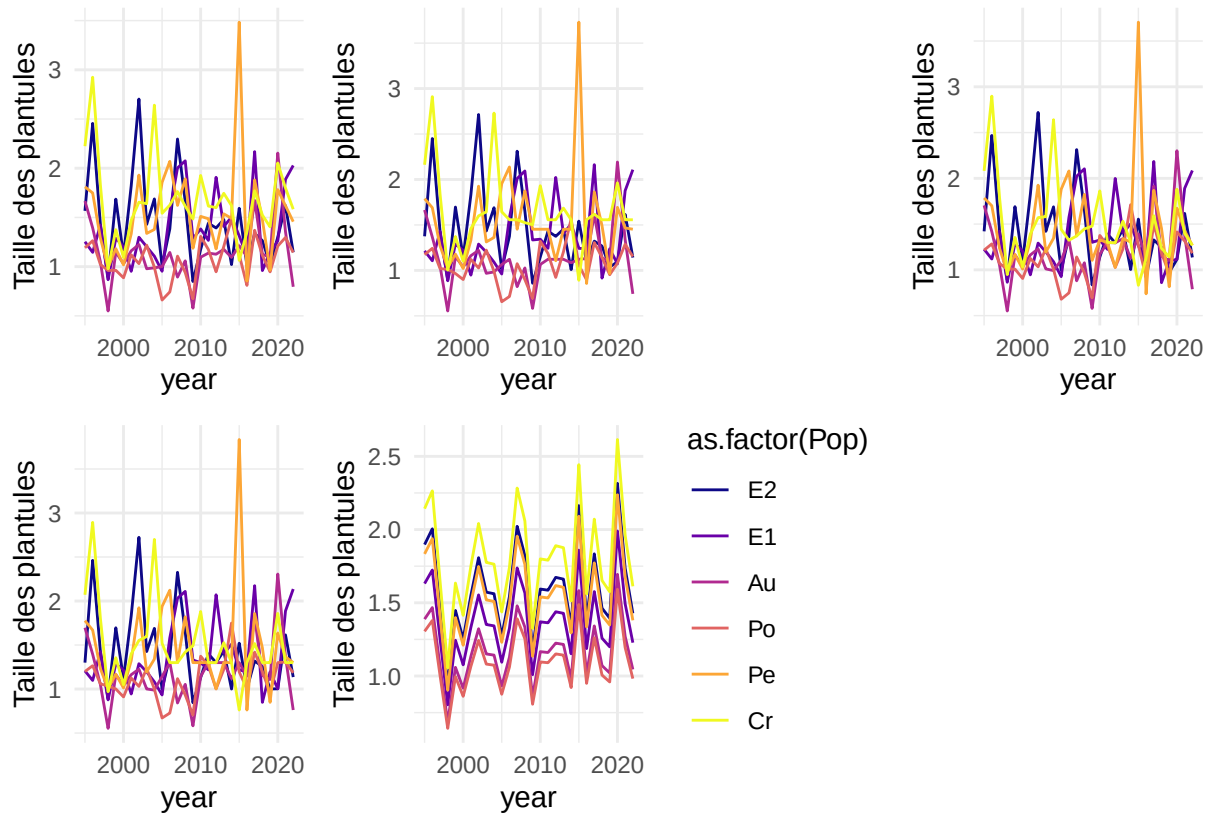


BIC

Seedling size

Same with AIC and BIC

As a function of year



Density

